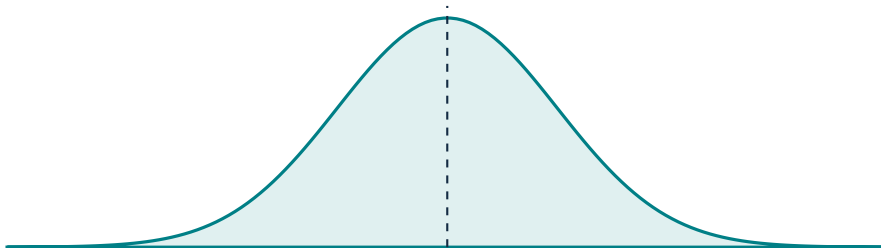


A FIRST-PRINCIPLES STUDY COMPANION

Probability and Statistics

MIT 18.05
Classes 1–10 and 22–23



Intuition before formulas. Proofs behind formulas.
Applications after formulas.

Original explanations, worked examples, practice problems,
complete exercise solutions, and a prerequisite bridge.

Aligned with the MIT OpenCourseWare Spring 2022 class numbering.
Independent learning material; not an official MIT publication.

How to use this course

This book is for someone who can manipulate equations but wants to know what the equations are *doing*. A formula is not the starting point. The starting point is an uncertain situation: a request might fail, a measurement might be noisy, a sample might mislead us. We then build the mathematical object that answers the question.

The selected MIT classes are a substantial part of 18.05, but not the entire semester. In the Spring 2022 numbering, class 8 is a review and class 9 is Exam 1. Here these become a structured review and an original practice exam. Class 10 introduces statistical inference and maximum likelihood. Classes 22–23 study confidence intervals. A clearly marked bridge supplies the sampling-distribution and testing background needed to make that jump.

The explanations and exercises are written for this companion. MIT’s official materials determine the scope and sequence, not the wording. Links to the source syllabus and readings are at the end. The book does not reproduce every MIT classroom question; it teaches the selected mathematical topics and adds its own practice. Further extensions are labeled so they do not get confused with the required class material.

The learning loop

For each class: read the motivating problem; explain the analogy aloud; derive the formula without looking; solve a worked example again; attempt the exercises; only then read their solutions. Finish by stating the assumptions that would make the method fail.

Prerequisites. Algebra, finite sums, basic differentiation and integration, and elementary double integrals. The optional CLT proof uses complex exponentials and a standard convergence theorem; those extra prerequisites are identified rather than concealed.

What “proof” means here. Elementary identities and theorems are proved at a calculus-based undergraduate level. The central limit theorem receives a characteristic-function proof with the underlying continuity theorem explicitly stated as an advanced imported result. The strong law is distinguished from the weak law but is not proved in its greatest generality. This avoids pretending that a first probability course contains a full course in measure theory.

A realistic study rhythm. Treat classes 5, 6, 7, 22, and 23 as more than one sitting. Deriving and solving is the work; reading quickly is not the same as learning. Use the review and exam to locate gaps before going on to inference.

Coverage map

MIT class	Topics developed here
1	Probability versus statistics; sets; product rule; permutations; combinations; inclusion-exclusion.
2	Experiments; sample spaces; probability functions; axioms; complements and unions; simulation; birthday collisions.
3	Conditional probability; multiplication and total probability; independence; Bayes; information protocols.

4	Discrete random variables; PMF and CDF; Bernoulli, binomial, geometric; expectation; linearity; indicators.
5	Variance and SD; continuous distributions; PDF and CDF; uniform, exponential, normal, Pareto; transformations.
6	Continuous moments and quantiles; sums and averages; histograms; LLN; Chebyshev; CLT and approximations.
7	Joint PMF/PDF/CDF; marginals; independence; covariance and correlation.
8	Integrated probability review and problem-selection strategies.
9	Original practice exam with worked solutions.
10	Models, parameters, statistics, likelihood, log-likelihood, maximum likelihood and its limitations.
Bridge	Sampling, sample variance, normal/chi-square/t distributions, testing, p-values, credible intervals.
22	Interval statistics; coverage; exact z, t, and chi-square confidence intervals; test inversion.
23	Pivotal, test-inversion, and formal views; non-normal means; Bernoulli proportions; conservative intervals; sample size.

Notation you must not mix up

Symbol	Meaning
X	Random numerical outcome, before observing it.
x	A realized value of X , after observing it.
$X_1, \dots, X_n$	The first, second, through n th observations; subscript n is an index, not a power.
$\bar{X}_n$	Random sample average, $n^{-1} \sum_i X_i$.
$\bar{x}$	Observed numerical average.
μ, σ^2	Population mean and variance.
S^2, s^2	Random sample variance and its observed value.
f, F	Density (or a stated PMF) and cumulative distribution function.
ϕ, Φ	Standard normal density and CDF.
$z_q, t_{q,\nu}, \chi_{q,\nu}^2$	Lower-tail q quantiles. Thus $z_{.975} \approx 1.96$.

Some MIT notes index critical values by *upper-tail* probability. This book consistently uses *lower-tail* quantiles. Our $z_{1-\alpha/2}$ is the positive critical value sometimes denoted $z_{\alpha/2}$ elsewhere.

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Class 1: counting and the language of events

1.1 Why this part exists

Imagine generating random identifiers for submissions on an online judge. You need to know how many possible identifiers exist and how likely two submissions are to receive the same one. Before calculating a probability, you must know what counts as a distinct outcome.

Counting is not a side topic. It is the engine behind finite probability. The same distinction you make in programming between an ordered vector and an unordered set decides whether a probability formula contains a factorial.

Probability goes forward; statistics goes backward

Probability: assuming a request fails with probability $p = .02$, how many failures might occur among 100 requests? Statistics: observing 3 failures among 100 requests, what can we infer about the unknown p ? The first starts with a model. The second tries to learn a model from data.

Frequentist reasoning evaluates procedures over repeated data sets with parameters held fixed. Bayesian reasoning represents uncertainty about parameters using probability distributions and updates them with data. Both use the same probability rules. The interpretation of what is random differs.

1.2 Sets: turning a sentence into an object

A sample space Ω contains possible outcomes. An event $A \subseteq \Omega$ contains the outcomes for which a statement is true. For one die, $\Omega = \{1, 2, 3, 4, 5, 6\}$. Let $A = \{2, 4, 6\}$ and $B = \{4, 5, 6\}$.

- $A \cap B = \{4, 6\}$ means both conditions hold.
- $A \cup B = \{2, 4, 5, 6\}$ means at least one holds; mathematical “or” includes both.
- $A^c = \{1, 3, 5\}$ means A does not occur.
- $A \setminus B = \{2\}$ means A occurs but B does not.
- A and B are disjoint if $A \cap B = \emptyset$.

The size of a finite set is its cardinality $|A|$. A subset relation is not a probability statement: $A \subseteq B$ says every outcome satisfying A also satisfies B .

Proposition 1.1 (De Morgan’s laws). $(A \cup B)^c = A^c \cap B^c$ and $(A \cap B)^c = A^c \cup B^c$.

Proof. An outcome fails to belong to the union exactly when it belongs to neither set. An outcome fails to belong to the intersection exactly when at least one membership condition fails. This proves both identities by testing membership outcome by outcome. $\square$

In real systems, “at least one server works” is the complement of “all servers fail.” Set language exposes the easier event to compute.

1.3 Counting rules and why they work

Product rule

If a task has a first choices and exactly b continuations for each first choice, there are ab complete outcomes. Think of a rooted decision tree with a branches, each with b leaves. With k stages and fixed branching counts $n_1, \dots, n_k$, the total is $\prod_i n_i$.

If different first choices have different numbers of continuations b_i , the answer is $\sum_i b_i$, not a blind product. In probability, equal branching counts still do not imply equal probabilities unless the mechanism gives that symmetry.

Permutations: positions are distinct

Ordering n distinct objects gives $n! = n(n-1) \cdots 1$ arrangements. Filling only r positions without replacement gives

$$P(n, r) = \frac{n!}{(n-r)!}.$$

The first position has n choices, the second $n-1$, and so on. Define $0! = 1$: there is one empty arrangement, not zero arrangements.

Combinations: positions are not distinct

Choosing an unordered r -element subset from n objects gives

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

Proof: count the ordered choices first, then divide by $r!$ because every selected subset was counted once for each internal ordering. Division by a factorial is a correction for repeated counting, not a magic ritual.

When n positions contain $n_1, \dots, n_k$ indistinguishable labels, with $\sum n_i = n$, the count is $n!/(n_1! \cdots n_k!)$. This will become the multinomial coefficient.

Addition and inclusion-exclusion

Disjoint alternatives add. For overlapping alternatives,

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

The overlap was counted twice; subtract it once. For three sets,

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

An outcome in all three sets is counted $3 - 3 + 1 = 1$ time. More generally, an outcome in exactly r sets receives total weight $\sum_{j=1}^r (-1)^{j+1} \binom{r}{j} = 1$, by expanding $(1-1)^r$. This proves general inclusion-exclusion.

Worked example: a contest team versus assigned roles

Choose three students from ten. If the only object is the team, there are $\binom{10}{3} = 120$ possibilities. If you choose a captain, coder, and analyst as distinct roles, there are $10 \cdot 9 \cdot 8 = 720$. Every team has $3! = 6$ role assignments, so $120 \cdot 6 = 720$.

Worked example: binary strings

There are $2^8 = 256$ strings of length eight. Exactly three ones means choosing their three positions: $\binom{8}{3} = 56$. Under independent fair bits, the probability is $56/256$. Under biased bits, the *count* stays 56, but each such string has probability $p^3(1-p)^5$.

Worked example: one pair in a five-card hand

Choose the pair's rank in 13 ways, its two suits in $\binom{4}{2}$ ways, three other distinct ranks in $\binom{12}{3}$ ways, and one suit for each in 4^3 ways. Thus

$$\mathbb{P}(\text{exactly one pair}) = \frac{13 \binom{4}{2} \binom{12}{3} 4^3}{\binom{52}{5}} \approx .4226.$$

Requiring the other three ranks to be distinct prevents accidentally counting two pairs or a full house. The denominator uses unordered hands because the numerator does too.

1.4 Recognizing the typical exercise

Use combinations for random committees or samples without roles; permutations for rankings or assignments; products for sequences; inclusion-exclusion for “at least one property” when categories overlap. Ask four questions before calculating: Does order matter? Are repetitions allowed? Are objects distinguishable? Are the counted outcomes equally likely?

1.5 Practice: try before reading the solutions

- 1.1 A code contains six decimal digits and may start with zero. Count codes with repetition and without repetition.
- 1.2 From 12 students, choose four, with at least one of two particular students included.
- 1.3 Count distinct arrangements of the letters in LEVEL.
- 1.4 Among 80 users, 45 use feature A, 35 use feature B, and 20 use both. How many use neither?
- 1.5 Prove $\binom{n}{r} = \binom{n-1}{r} + \binom{n-1}{r-1}$ without factorial manipulation.
- 1.6 A random five-element subset of $\{1, \dots, 20\}$ is selected. Find the probability that exactly two selected numbers are even.

1.6 Solutions and reasoning

- 1.1 With repetitions, each slot has ten choices, so 10^6 . Without them, $10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 = 151200$. Leading zero is allowed, so the first factor remains ten.

- 1.2 Subtract teams containing neither: $\binom{12}{4} - \binom{10}{4} = 495 - 210 = 285$. This avoids separately counting teams containing both special students.
- 1.3 Five positions, with two L's and two E's: $5!/(2!2!) = 30$.
- 1.4 The union contains $45 + 35 - 20 = 60$, leaving 20. Adding 45 and 35 without subtracting the overlap would count shared users twice.
- 1.5 Fix one distinguished person. Every r -person team either excludes that person, contributing $\binom{n-1}{r}$, or includes them, contributing $\binom{n-1}{r-1}$. The cases partition all teams.
- 1.6 There are ten evens and ten odds. Favorable subsets number $\binom{10}{2}\binom{10}{3}$. Divide by $\binom{20}{5}$, giving about .3483.

Chapter 2

Class 2: probability as a mathematical model

2.1 Why this part exists

A random experiment is not described fully by listing possible outcomes. A server can succeed or fail, but that does not make failure a 50% event. We need a rule assigning probability to outcomes and events.

Think of probability as a total budget of one unit of mass. Events collect some of that mass. A calculation can be perfectly correct inside a model while the model itself is unrealistic. Separating those two questions is a central scientific habit.

2.2 Experiments, outcomes, and probability functions

For two fair independent dice, use ordered pairs (i, j) with $1 \leq i, j \leq 6$. There are 36 equally likely outcomes. The eleven possible sums $2, \dots, 12$ are *not* equally likely: sum 2 has one pair, while sum 7 has six.

For a finite sample space, assign masses $p(\omega) \geq 0$ summing to one and define

$$\mathbb{P}(A) = \sum_{\omega \in A} p(\omega).$$

Only in the uniform finite case does this reduce to $|A|/|\Omega|$.

The general probability axioms are nonnegativity, $\mathbb{P}(\Omega) = 1$, and countable additivity: for pairwise disjoint events A_i ,

$$\mathbb{P}\left(\bigcup_i A_i\right) = \sum_i \mathbb{P}(A_i).$$

In full measure theory probabilities are defined on a suitable collection of events called a sigma-algebra. Finite spaces need no such complication: every subset is allowed. The continuous models here use the usual measurable events.

2.3 Rules derived from the axioms

Because A and A^c partition Ω , $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$. The empty event has probability zero. If $A \subseteq B$, write $B = A \cup (B \setminus A)$ as a disjoint union; hence $\mathbb{P}(A) \leq \mathbb{P}(B)$.

Decompose $A \cup B$ into $A \setminus B$, $B \setminus A$, and $A \cap B$. Adding their probabilities gives

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

Since the subtracted term is nonnegative,

$$\mathbb{P}\left(\bigcup_{i=1}^m A_i\right) \leq \sum_{i=1}^m \mathbb{P}(A_i).$$

This is the union bound. Repeated application proves it for finite m ; countable versions follow from countable additivity. It needs no independence and is useful for bounding the chance that *any* of many possible failures happens.

Worked example: reliability without assuming independence

Two components fail with probabilities .03 and .04. If either failure stops the system, the system failure probability is at most .07. It is at least .04, because the event includes failure of the second component. You cannot claim the exact value $.03 + .04 - .03 \cdot .04$ without independence.

Worked example: birthday matches and hash collisions

Suppose m people have independent birthdays uniformly distributed over $d = 365$ days, ignoring leap years. Rather than count all ways to have matches, count no match:

$$\mathbb{P}(\text{no collision}) = \prod_{j=0}^{m-1} \left(1 - \frac{j}{d}\right), \quad m \leq d.$$

Each arriving person must avoid already occupied days. The collision probability is one minus this product. For $m = 23$ it is about .5073.

Why does the threshold seem surprisingly small? There are $\binom{m}{2}$ pairs, not just m opportunities. Using $\log(1 - u) \approx -u$ for small u gives

$$\mathbb{P}(\text{collision}) \approx 1 - \exp\left(-\frac{m(m-1)}{2d}\right).$$

For uniform random identifiers from d possibilities, exactly the same reasoning applies. The rough collision scale is $\sqrt{d}$, not d . The pair events are not jointly independent; this approximation does not assert that they are.

2.4 Simulation is an experiment about the model

To estimate a probability, simulate the experiment N times and average an indicator that equals one when the target event occurs. If the event probability is p , the estimator has mean p and, for independent simulations, variance $p(1 - p)/N$ (proved in later classes).

Doubling the number of trials does not halve the standard error; multiplying trials by four does. Simulation is a useful numerical check, but it does not prove a theorem and cannot validate a flawed real-world model merely by agreeing with its formulas.

Zero probability and impossibility are different in continuous models

On a finite space with every point assigned positive probability, probability zero means no allowed outcome is in the event. In a continuous model, an exact point can have probability zero even though the observation must land at some point. We will resolve this through density and integrals in class 5.

2.5 Practice: try before reading the solutions

- 2.1 Two fair independent dice are rolled. Find the probabilities that their sum is 9 and that their sum is at least 11.

- 2.2 $\mathbb{P}(A) = .6$, $\mathbb{P}(B) = .5$, and $\mathbb{P}(A \cap B) = .3$. Find union, neither, and exactly one.
- 2.3 One hundred components each fail with probability .001. Give an upper bound on at least one failure, without independence.
- 2.4 Four random identifiers are independently uniform among ten values. Find the collision probability.
- 2.5 A probability estimate from 10000 independent simulations equals .2. Estimate its standard error. How many trials reduce that standard error by a factor of three?
- 2.6 If $\mathbb{P}(A) = .8$ and $\mathbb{P}(B) = .7$, prove that $\mathbb{P}(A \cap B) \geq .5$.

2.6 Solutions and reasoning

- 2.1 Sum 9 occurs at $(3, 6), (4, 5), (5, 4), (6, 3)$, giving $4/36 = 1/9$. Sums 11 and 12 have two and one pairs, giving $3/36 = 1/12$.
- 2.2 Union $.6 + .5 - .3 = .8$; neither .2; exactly one $\mathbb{P}(A \cup B) - \mathbb{P}(A \cap B) = .5$.
- 2.3 The union bound gives .1. This is a bound, not an exact probability.
- 2.4 No collision has probability $1 \cdot .9 \cdot .8 \cdot .7 = .504$. Collision probability is .496.
- 2.5 Plug $\hat{p} = .2$ into $\sqrt{\hat{p}(1 - \hat{p})/N}$: .004. Reducing by three requires multiplying N by nine, to 90000. This estimates simulation uncertainty, not model error.
- 2.6 Inclusion-exclusion and $\mathbb{P}(A \cup B) \leq 1$ give $\mathbb{P}(A \cap B) \geq \mathbb{P}(A) + \mathbb{P}(B) - 1 = .5$.

Class 3: learning from information

3.1 Why this part exists

An alarm sounds. Does that mean a server is actually down? Knowing how often an alarm fires *when a server is down* is not enough. We also need how common outages are and how often alarms fire when nothing is wrong.

Conditioning is the operation that changes the population you are looking at. Bayes' theorem is a bookkeeping rule for reversing the direction of that information.

3.2 Conditional probability: restrict, then renormalize

If $\mathbb{P}(B) > 0$, define

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

The denominator says: among the probability mass that survives the information B , what fraction also satisfies A ? It is like applying a database filter and recomputing percentages on the filtered rows.

For two dice, given that the first die is 4, the chance of sum 9 is $1/6$. Given that the sum is 9, the chance that the first die is 4 is $1/4$. Reversing a conditional changes the denominator and usually changes the answer.

Rearranging the definition gives the multiplication rule,

$$\mathbb{P}(A \cap B) = \mathbb{P}(B)\mathbb{P}(A | B).$$

Repeatedly applying it gives the chain rule

$$\mathbb{P}(A_1 \cap \dots \cap A_k) = \mathbb{P}(A_1) \prod_{j=2}^k \mathbb{P}(A_j | A_1 \cap \dots \cap A_{j-1}),$$

where the conditioning events must have positive probability for the displayed ratios to be defined.

3.3 Independence is an assumption about a relationship

Events A, B are independent if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. If $\mathbb{P}(B) > 0$, this is equivalent to $\mathbb{P}(A | B) = \mathbb{P}(A)$. Knowing B does not change the probability of A .

Disjoint events with positive probabilities cannot be independent: their intersection has probability zero, but the product of their probabilities is positive. Disjoint means “cannot happen together”; independent means “learning one gives no information about the other.”

Proposition 3.1. *If A and B are independent, then A and B^c are independent.*

Proof. $\mathbb{P}(A \cap B^c) = \mathbb{P}(A) - \mathbb{P}(A \cap B) = \mathbb{P}(A)[1 - \mathbb{P}(B)] = \mathbb{P}(A)\mathbb{P}(B^c)$. □

For several events, *mutual* independence requires product factorization for every subcollection, not merely for pairs. Let U, V be independent fair bits and let $W = U \text{ xor } V$. Every pair is independent, but U, V determine W . Three-way independence fails.

Conditional independence given C is the analogous factorization inside the C world. It need not imply unconditional independence. Two servers can fail independently given today's power condition but be dependent overall because they share the same power supply.

3.4 Total probability and Bayes' theorem

Suppose $H_1, \dots, H_k$ partition the sample space. Since the events $B \cap H_i$ are disjoint,

$$\mathbb{P}(B) = \sum_i \mathbb{P}(B | H_i) \mathbb{P}(H_i).$$

This is the law of total probability: add the probability of each route into the observed event. Combining it with the multiplication rule proves

$$\mathbb{P}(H_j | B) = \frac{\mathbb{P}(B | H_j) \mathbb{P}(H_j)}{\sum_i \mathbb{P}(B | H_i) \mathbb{P}(H_i)}.$$

The prior $\mathbb{P}(H_j)$ is the weight before the new information; the likelihood $\mathbb{P}(B | H_j)$ measures how compatible the observation is with that hypothesis; the posterior is the updated weight.

Worked example: an outage alarm and the base rate

Suppose a randomly selected server-minute has an outage with probability .01. The alarm detects 95% of outages and falsely fires in 4% of non-outage minutes. Then

$$\mathbb{P}(\text{outage} | \text{alarm}) = \frac{.95(.01)}{.95(.01) + .04(.99)} \approx .1935.$$

In 10000 model-generated server-minutes, expect 100 outage minutes and 9900 healthy ones. About 95 alarms come from outages and 396 from healthy minutes. Of 491 alarms, only 95 are true outages. This is not a defect in Bayes' formula; it is the consequence of a rare event competing with many opportunities for false alarms.

Worked example: sampling without replacement

A bag contains six red and four blue balls. The probability of drawing two red balls is $\frac{6}{10} \frac{5}{9} = \frac{1}{3}$. The second probability changes after the first draw. Drawing with replacement would instead give $.6^2 = .36$. The wording determines the dependence structure.

Worked example: why the information protocol matters

In the standard three-door game, the host knows the prize location, always reveals an unchosen losing door, and always offers a switch. Initially your door wins with probability $1/3$. Switching wins exactly when the original choice was wrong, probability $2/3$.

If instead someone randomly opens an unchosen door and happens not to reveal the prize, that is a different information mechanism. Condition on surviving: initial correctness contributes $(1/3) \cdot 1$ and initial incorrectness contributes $(2/3) \cdot (1/2)$. They are equal, so switching then wins with conditional probability $1/2$. Never erase the process that produced the information.

3.5 Where this is used

Debugging from symptoms, spam classification, quality-control flags, reliability trees, and interpreting positive tests are reverse-conditional problems. A typical exercise supplies $\mathbb{P}(\text{evidence} \mid \text{cause})$ and asks for $\mathbb{P}(\text{cause} \mid \text{evidence})$. Another typical exercise checks whether the experiment samples with or without replacement.

3.6 Practice: try before reading the solutions

- 3.1 Supplier A provides 70% of parts with 1% defects; B provides 30% with 5% defects. A part is defective. Find the probability it came from B.
- 3.2 A fair coin is tossed twice. Let A be first toss heads, B second toss heads, and C exactly one head. Check pairwise and mutual independence.
- 3.3 From five working and three faulty devices, choose two without replacement. Given at least one is faulty, find the probability both are faulty.
- 3.4 Prove that independence of A, B implies independence of A^c, B^c .
- 3.5 A hypothesis has prior odds 1 : 9. An observation is six times as likely under it as under its complement. Find posterior odds and probability.
- 3.6 In the alarm example, why is it not generally valid to square both likelihoods for two repeated alarms from the same sensor?

3.7 Solutions and reasoning

- 3.1 Total defect probability is $.7(.01) + .3(.05) = .022$. The B-and-defect probability is .015, so posterior $.015/.022 = 15/22 \approx .6818$.
- 3.2 Each event has probability $1/2$. Each pair intersection has probability $1/4$: $A \cap B = \{HH\}$, $A \cap C = \{HT\}$, $B \cap C = \{TH\}$. But the triple intersection is empty rather than probability $1/8$. Pairwise, not mutual, independence holds.
- 3.3 Both faulty has probability $\binom{3}{2}/\binom{8}{2} = 3/28$. At least one faulty has probability $1 - \binom{5}{2}/\binom{8}{2} = 18/28$. The conditional answer is $1/6$.
- 3.4 $\mathbb{P}(A^c \cap B^c) = 1 - \mathbb{P}(A \cup B) = 1 - \mathbb{P}(A) - \mathbb{P}(B) + \mathbb{P}(A)\mathbb{P}(B) = (1 - \mathbb{P}(A))(1 - \mathbb{P}(B))$.
- 3.5 Bayes in odds form multiplies prior odds by likelihood ratio: posterior odds $6 : 9 = 2 : 3$, hence posterior probability $2/5$. Derive the rule by dividing Bayes' formulas for H and H^c .
- 3.6 Squaring assumes the alarms are conditionally independent given outage and also given no outage. A stuck sensor or shared noise can make repeated alarms strongly dependent. Repeating evidence is not necessarily obtaining new evidence.

Class 4: random variables and expectation

4.1 Why this part exists

A full outcome can contain too much detail. For 100 requests, the outcome is an entire success/failure sequence. Often all we care about is the number of failures, the time to the first failure, or the cost. A random variable extracts that numerical feature.

A random variable is a function, not a mysterious moving number

$X : \Omega \rightarrow \mathbb{R}$ assigns a number to each outcome. The randomness lies in which ω occurs. Once ω is fixed, $X(\omega)$ is fixed. A distribution describes how the outcome probabilities are collected by that function.

4.2 PMF, CDF, and support

For a discrete random variable, its probability mass function is $p_X(x) = \mathbb{P}(X = x)$. Its support is the set of values with positive mass. A valid PMF has nonnegative masses summing to one.

The cumulative distribution function is

$$F_X(t) = \mathbb{P}(X \leq t) = \sum_{x \leq t} p_X(x).$$

It is nondecreasing, right-continuous, and tends to zero and one at the two infinities. A jump at x has size $p_X(x)$. Right-continuity follows because when $t_m \downarrow t$, the events $\{X \leq t_m\}$ decrease to $\{X \leq t\}$, and probabilities are continuous under such decreasing limits.

Always use $\mathbb{P}(a < X \leq b) = F_X(b) - F_X(a)$. For an integer-valued X , $\mathbb{P}(a \leq X \leq b) = F_X(b) - F_X(a - 1)$ when a, b are integers. Endpoint details matter for discrete variables.

4.3 Three central discrete families

Bernoulli: one yes/no experiment

$X \sim \text{Bern}(p)$ means $\mathbb{P}(X = 1) = p$ and $\mathbb{P}(X = 0) = 1 - p$. A request failure indicator, whether a sampled voter supports an option, or whether a solution passes a test can all be Bernoulli variables. The application may be complex; the recorded observation has two values.

Binomial: count successes in a fixed number of trials

For n independent Bernoulli trials with the same p , their sum K obeys

$$\mathbb{P}(K = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, \dots, n.$$

Each success-position pattern has probability $p^k(1-p)^{n-k}$, and there are $\binom{n}{k}$ such patterns. The binomial theorem shows the probabilities sum to $(p + 1 - p)^n = 1$.

The requirements are a fixed number of trials, two outcomes per trial, independence, and a common success probability. A sample without replacement is generally not binomial. Nor is a sequence with changing success probabilities, although other models can handle both.

Geometric: wait for the first success

Let T count trials *including* the first success. For $0 < p \leq 1$,

$$\mathbb{P}(T = k) = (1-p)^{k-1}p, \quad k = 1, 2, \dots; \quad \mathbb{P}(T > k) = (1-p)^k.$$

The first $k-1$ trials fail and the k th succeeds. Summing the geometric series normalizes the PMF. Its memoryless property is

$$\mathbb{P}(T > s+t \mid T > s) = (1-p)^t = \mathbb{P}(T > t),$$

for nonnegative integers s, t . Previous failed independent attempts do not make the next attempt more likely to succeed. Some texts count failures before success instead: $G = T - 1$, with support starting at zero. Check the convention.

4.4 Expectation: the center of probability mass

When $\sum_x |x|p_X(x) < \infty$, define

$$\mathbb{E}[X] = \sum_x xp_X(x).$$

Imagine placing masses $p_X(x)$ along a ruler at coordinates x . The expected value is the balance point. It need not be an outcome: a fair die has mean 3.5, although no face says 3.5. It need not be the most likely outcome, nor the median.

For a function g , the law of the unconscious statistician gives

$$\mathbb{E}[g(X)] = \sum_x g(x)p_X(x),$$

assuming integrability. Proof: group outcomes by the value of $g(X)$ and then regroup by X ; all mass mapping to a particular output receives that output's value. This avoids first computing the entire distribution of $g(X)$.

Theorem 4.1 (Linearity). *For integrable X, Y and constants a, b , $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$, regardless of dependence.*

Proof. Sum over underlying outcomes: $\sum_\omega [aX(\omega) + bY(\omega)]p(\omega)$ splits into two sums. The same reasoning works using joint distributions or integrals. Absolute integrability justifies regrouping in infinite cases. $\square$

Linearity is often the shortcut that removes the need to find a complicated distribution.

Means of the three families

For Bernoulli, $\mathbb{E}[X] = p$. A binomial is a sum of n indicators, so $\mathbb{E}[K] = np$.

For geometric, put $q = 1 - p$ and differentiate $\sum_{k=0}^{\infty} q^k = 1/(1 - q)$ for $|q| < 1$:

$$\mathbb{E}[T] = p \sum_{k=1}^{\infty} k q^{k-1} = \frac{p}{(1 - q)^2} = \frac{1}{p}.$$

Alternatively, after one trial you either succeed or restart: $m = 1 + (1 - p)m$, giving $m = 1/p$. The series computation also establishes finiteness and justifies the restart equation.

Worked example: retrying a randomized procedure

An independent attempt succeeds with probability .2. The expected number of attempts is five, and the probability of needing more than ten attempts is $.8^{10} \approx .1074$. An expectation is not a deadline guarantee. If each attempt costs 3 ms, expected total cost is 15 ms.

Worked example: fixed points in a random permutation

Shuffle n identifiers uniformly. Let $I_i = 1$ if item i returns to its original position. Although these events are dependent, $\mathbb{E}[I_i] = 1/n$, so

$$\mathbb{E} \left[\sum_{i=1}^n I_i \right] = n \cdot \frac{1}{n} = 1.$$

One expected fixed point for every n is a clean example of why expectation can be easier than the full distribution.

Worked example: expected number of colliding pairs

Among m uniform independent identifiers in a space of size d , let I_{ij} indicate that items i, j match. Each pair matches with probability $1/d$, so the expected number of colliding pairs is $\binom{m}{2}/d$. This is not the probability of any collision: one experiment can contain several colliding pairs. However, the union bound gives $\mathbb{P}(\text{any collision}) \leq \binom{m}{2}/d$.

4.5 Expectation can fail to exist

If positive and negative contributions both diverge, one may not cancel them informally. Even for nonnegative variables, the mean can be infinite. We will see heavy-tailed distributions for which collecting more observations does not produce the usual finite-mean behavior. “Every distribution has an average” is false.

4.6 Practice: try before reading the solutions

4.1 X takes values 0, 1, 3 with probabilities .2, .5, .3. Find its CDF, $\mathbb{E}[X]$, and $\mathbb{E}[X^2]$.

4.2 Each of 20 independent requests fails with probability .1. Find expected failures and the probability of exactly two failures.

- 4.3 An attempt succeeds with probability $1/4$. Find the probability the first success is on attempt 4, and the expected number of failures before success.
- 4.4 A payout is 100 DH with probability .03 and zero otherwise. What entry fee gives zero expected net profit? Does that make one play low-risk?
- 4.5 Randomly permute eight items. Find the expected number not in their original places.
- 4.6 Prove $\mathbb{E}[T] = \sum_{k=0}^{\infty} \mathbb{P}(T > k)$ for any nonnegative integer-valued T (allowing infinity), then apply it to geometric T .

4.7 Solutions and reasoning

- 4.1 $F(t) = 0$ for $t < 0$, .2 for $0 \leq t < 1$, .7 for $1 \leq t < 3$, and 1 for $t \geq 3$. The mean is $.5 + .9 = 1.4$ and second moment $.5 + 2.7 = 3.2$.
- 4.2 $K \sim \text{Bin}(20, .1)$, so $\mathbb{E}[K] = 2$ and $\mathbb{P}(K = 2) = \binom{20}{2}(.1)^2(.9)^{18} \approx .2852$.
- 4.3 The probability is $(3/4)^3(1/4) = 27/256$. The mean trial count is 4, so the mean failure count is 3. This explicitly distinguishes the two geometric conventions.
- 4.4 The expected payout is 3 DH, so the fee is 3 DH. In one play the net is usually -3 and occasionally 97: zero mean says nothing by itself about small variability or personal utility.
- 4.5 Expected fixed points is 1. By linearity, expected non-fixed points is $8 - 1 = 7$.
- 4.6 Pointwise $T = \sum_{k=0}^{\infty} \mathbf{1}_{\{T > k\}}$. Taking expectations and interchanging a nonnegative sum and expectation gives the identity. For geometric T , sum $(1 - p)^k$ to obtain $1/p$.

Class 5: variability, densities, and transformations

5.1 Why this part exists

Two algorithms can have the same average runtime but very different reliability. One always takes 10 ms; another takes 1 ms most of the time but occasionally takes hundreds. The average alone cannot answer a timeout question. We need a measure of spread and, beyond that, the full distribution.

Continuous quantities raise another difficulty. If time can take any real value in an interval, we cannot assign a positive mass to every point and keep total probability one. Density solves this by assigning probability to *intervals through area*.

5.2 Variance and standard deviation

For a finite second moment, define

$$\mu = \mathbb{E}[X], \quad \text{Var}(X) = \mathbb{E}[(X - \mu)^2], \quad \sigma = \sqrt{\text{Var}(X)}.$$

Signed deviations would average to zero. Squaring prevents cancellation and penalizes large deviations. Variance has squared units; SD returns to the original units. Squared error is not the only possible loss, but it has exceptionally useful algebra.

Proposition 5.1 (Computational identities).

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2, \quad \text{Var}(aX + b) = a^2 \text{Var}(X).$$

Proof. Expand $(X - \mu)^2 = X^2 - 2\mu X + \mu^2$ and take expectations. For the second formula, $aX + b - \mathbb{E}[aX + b] = a(X - \mu)$; square and take expectations. Therefore $\text{SD}(aX + b) = |a|\sigma$. $\square$

Adding 100 ms to every runtime changes the mean, not the spread. Changing from seconds to milliseconds multiplies the SD by 1000 and the variance by one million.

For independent finite-variance variables,

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$

Proof: the cross term has expectation $2\mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)] = 0$ because independence factors it into two zero means. Without independence the cross term need not vanish; class 7 names it covariance.

Discrete family variances

For Bernoulli X , $X^2 = X$, so $\text{Var}(X) = p - p^2 = p(1 - p)$. For binomial $K = \sum_i X_i$, independence gives $\text{Var}(K) = np(1 - p)$.

For geometric T on $1, 2, \dots$, differentiating the geometric series twice gives

$$\sum_{k=1}^{\infty} k(k-1)q^{k-2} = \frac{2}{(1-q)^3}.$$

Thus $\mathbb{E}[T(T-1)] = 2q/p^2$, and

$$\mathbb{E}[T^2] = \frac{2-p}{p^2}, \quad \text{Var}(T) = \frac{1-p}{p^2}.$$

When success is rare, geometric waiting times have both a large mean and a large spread.

Worked example: same mean, different behavior

Let $A = 10$ with certainty, while $B = 0$ or 20 with equal probabilities. Both means are 10, but $\text{Var}(A) = 0$ and $\text{Var}(B) = 100$. A deadline of 15 is always met by A and met only half the time by B . Mean optimization and tail-risk optimization are distinct goals.

5.3 From probability mass to probability density

For an absolutely continuous random variable there is a nonnegative function f with

$$\int_{-\infty}^{\infty} f(x) dx = 1, \quad \mathbb{P}(a < X \leq b) = \int_a^b f(x) dx.$$

The PDF value $f(x)$ is not $\mathbb{P}(X = x)$. It has units inverse to those of x : if x is time in seconds, density has units per second. Multiplying by a small width gives approximately a probability:

$$\mathbb{P}(x < X \leq x+h) \approx f(x)h$$

when h is small and the density varies little there.

Density is concentration, not probability at a point

Think of one kilogram of sand spread along a line. Height represents mass per unit length. A thin interval contains height times width; a point has zero width. A tall pile can have height greater than one without containing more than one kilogram in total.

For $X \sim \text{Unif}(0, .1)$ the density is 10, but the probability of an interval of width .02 inside the support is only .2. At each exact point the probability is zero. Probability is countably additive, not a rule permitting an uncountable sum of point masses to be treated as a countable sum.

We use “continuous” in this course for models with a density. In advanced probability there are also continuous CDFs without densities, and distributions mixing point masses with continuous parts.

5.4 The CDF unifies discrete and continuous probability

In every model, $F(x) = \mathbb{P}(X \leq x)$. With a density,

$$F(x) = \int_{-\infty}^x f(t) dt, \quad f(x) = F'(x)$$

at points where the derivative identity holds. The fundamental theorem of calculus gives the second equation when f is continuous there. Endpoints no longer change probabilities because singletons have zero probability.

Worked example: normalize first, then compute

Suppose $f(x) = cx$ for $0 < x < 2$ and zero elsewhere. Normalization gives $1 = \int_0^2 cx \, dx = 2c$, so $c = 1/2$. Then

$$F(x) = \begin{cases} 0 & x \leq 0, \\ x^2/4 & 0 < x < 2, \\ 1 & x \geq 2. \end{cases}$$

The probability of $1 < X < 1.5$ is $F(1.5) - F(1) = .5625 - .25 = .3125$. The density is not flat: equal-width intervals nearer 2 contain more probability.

5.5 A gallery of models: what mechanism suggests which family?**Uniform: constant concentration on a bounded interval**

For $a < b$, $X \sim \text{Unif}(a, b)$ has

$$f(x) = \frac{1}{b-a} \quad (a < x < b), \quad F(x) = \frac{x-a}{b-a} \quad (a \leq x \leq b),$$

with CDF zero below a and one above b . Equal-length subintervals have equal probability. Examples include an ideal uniform random number and a rounding error when the fractional part is approximately uniform.

If a train comes exactly every 12 minutes and your arrival phase is uniform and independent of the schedule, the wait is uniform on $(0, 12)$. It is not exponential: having already waited 11 minutes tells you a great deal.

Exponential: a constant instantaneous event rate

For rate $\lambda > 0$,

$$f(x) = \lambda e^{-\lambda x} \quad (x \geq 0), \quad F(x) = 1 - e^{-\lambda x}, \quad \mathbb{P}(X > x) = e^{-\lambda x}.$$

Suppose arrivals occur at a constant rate with independent increments, as in an ideal Poisson process. Then the waiting time to the next arrival is exponential. The units of λ are events per unit time; the mean wait is $1/\lambda$. Do not invert a rate twice.

One derivation starts with the survival function $S(t) = \mathbb{P}(X > t)$. A constant small-interval hazard means $S(t+h) \approx S(t)(1-\lambda h)$, so $S'(t) = -\lambda S(t)$ with $S(0) = 1$. Solving gives $S(t) = e^{-\lambda t}$ and $f = -S'$.

Memorylessness follows directly:

$$\mathbb{P}(X > s+t \mid X > s) = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t}.$$

Actual component lifetimes need not be exponential: aging changes the hazard. An exponential is a mechanism assumption, not a universal law for waiting times.

Worked example: request arrivals

Under a homogeneous Poisson-arrival model with rate 3 per second, the chance of waiting more than half a second is $e^{-1.5} \approx .2231$. The chance of an arrival within .1 second is $1 - e^{-.3} \approx .2592$. These are model calculations, not promises about bursty production traffic.

Normal: a location-scale bell shape

For $\sigma > 0$,

$$X \sim \mathcal{N}(\mu, \sigma^2), \quad f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp \left[-\frac{(x - \mu)^2}{2\sigma^2} \right].$$

The notation's second parameter is *variance*. Many software functions instead accept SD. The curve centers at μ and spreads on scale σ . Increasing σ makes it wider and lower, preserving total area one.

Small additive measurement errors sometimes suggest a normal model; sums and averages explain its broader importance in class 6. Not every natural quantity is normal. Counts, bounded proportions, and highly skewed durations may need different models.

For $Z \sim \mathcal{N}(0, 1)$ write its density as $\phi(z)$ and CDF as $\Phi(z)$. Symmetry gives $\Phi(-z) = 1 - \Phi(z)$. Approximately 68.27%, 95.45%, and 99.73% lie within 1, 2, and 3 SDs. A central 95% interval uses 1.96 rather than exactly 2.

Why the normalizing constant is what it is

Let $I = \int_{-\infty}^{\infty} e^{-x^2/2} dx$. Square it and use polar coordinates:

$$I^2 = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)/2} dx dy = \int_0^{2\pi} \int_0^{\infty} e^{-r^2/2} r dr d\theta = 2\pi.$$

Since $I > 0$, $I = \sqrt{2\pi}$. Substituting $z = (x - \mu)/\sigma$ explains both the σ and $\sqrt{2\pi}$ factors. The CDF has no elementary antiderivative, but its integral definition is exact and numerical evaluation is routine.

Pareto: a heavy-tail warning and useful extension

For minimum $m > 0$ and tail parameter $\alpha > 0$,

$$f(x) = \frac{\alpha m^\alpha}{x^{\alpha+1}}, \quad F(x) = 1 - \left(\frac{m}{x}\right)^\alpha, \quad x \geq m.$$

The survival tail decays as a power, much more slowly than an exponential. It is a candidate model for certain size distributions and an instructive warning about extremes. It should be fitted and checked, not assumed merely because data look unequal.

For $r > 0$, direct integration gives

$$\mathbb{E}[X^r] = \alpha m^\alpha \int_m^\infty x^{r-\alpha-1} dx = \frac{\alpha m^r}{\alpha - r} \quad \text{if } \alpha > r,$$

and infinity otherwise. Thus the mean is $\alpha m/(\alpha - 1)$ for $\alpha > 1$, and variance is $\alpha m^2/[(\alpha - 1)^2(\alpha - 2)]$ for $\alpha > 2$. A finite mean does not guarantee finite variance.

5.6 Transforming random variables

If $Y = g(X)$, the safest method is to translate $Y \leq y$ back into a condition on X :

$$F_Y(y) = \mathbb{P}(g(X) \leq y).$$

For a strictly increasing differentiable g with nonzero derivative and inverse h , $F_Y(y) = F_X(h(y))$, so

$$f_Y(y) = f_X(h(y))|h'(y)|.$$

For decreasing g , the inequality reverses, and differentiation supplies the same absolute-value formula. For several inverse branches, sum their contributions. The derivative factor preserves probability mass when stretching or compressing the horizontal axis.

Worked example: standardization is a reversible change of units

For any X with finite mean and positive finite SD, $Z = (X - \mu)/\sigma$ has mean zero and variance one. If X is normal, substituting into the density shows that Z is standard normal. If X is not normal, standardization does not make it normal.

No information is lost if μ and σ are retained: $X = \mu + \sigma Z$. You have changed from “milliseconds” to “standard deviations relative to the mean.” If you discard the transformation parameters, you discard location and scale information.

If latency is modeled as $\mathcal{N}(100, 15^2)$, then $\mathbb{P}(X > 130) = \mathbb{P}(Z > 2) \approx .0228$. For an interval, transform *both* endpoints before subtracting CDF values.

Worked example: a non-monotone transformation

Let $X \sim \text{Unif}(-1, 1)$ and $Y = X^2$. For $0 \leq y \leq 1$,

$$F_Y(y) = \mathbb{P}(-\sqrt{y} \leq X \leq \sqrt{y}) = \sqrt{y}, \quad f_Y(y) = \frac{1}{2\sqrt{y}} \quad (0 < y < 1).$$

The density is unbounded near zero but integrates to one. Two X values, $\sqrt{y}$ and $-\sqrt{y}$, contribute. Missing one branch gives half the correct mass.

5.7 Extra intuition: how a sampler uses the CDF

A sampler is an algorithm producing values with a desired distribution. Let $U \sim \text{Unif}(0, 1)$. For a continuous strictly increasing CDF F , define $X = F^{-1}(U)$. Then

$$\mathbb{P}(X \leq x) = \mathbb{P}(U \leq F(x)) = F(x).$$

The construction works because a uniform random number chooses a percentile. The inverse CDF tells you which value occupies that percentile. You are not flipping the PDF's axes.

For a general CDF use the generalized inverse $F^{-1}(u) = \inf\{x : F(x) \geq u\}$. For Bernoulli, a threshold rule such as $X = \mathbf{1}_{\{U < p\}}$ works. For uniform, $X = a + (b - a)U$. For exponential,

$$X = -\frac{\log(1 - U)}{\lambda}.$$

Since $1 - U$ is also uniform, $-\log U/\lambda$ is equivalent in distribution. Implementations must handle endpoints to avoid $\log 0$. For normal and Student t , inverse CDFs may require numerical methods, or one can use other exact mathematical constructions. There is no requirement that every sampler use an elementary inverse formula.

5.8 Practice: try before reading the solutions

- 5.1 For the discrete X from exercise 4.1, compute variance and the variance of $3X - 2$.
- 5.2 $f(x) = cx^2$ for $0 < x < 1$. Find c , the CDF, and $\mathbb{P}(X > .8)$.
- 5.3 A uniform wait lies between zero and 15 minutes. Given you have already waited 10 minutes, what is the chance of waiting at least 3 more?
- 5.4 For an exponential lifetime of mean 8 hours, find $\mathbb{P}(X > 12 \mid X > 5)$ and the median lifetime.
- 5.5 $X \sim \mathcal{N}(50, 4^2)$. Find $\mathbb{P}(46 < X < 58)$ and its 90th percentile, using $z_{.90} = 1.2816$.
- 5.6 $X \sim \text{Unif}(0, 1)$ and $Y = -\log X$. Derive the CDF and identify the distribution.
- 5.7 $X \sim \mathcal{N}(0, 1)$ and $Y = X^2$. Derive $f_Y(y)$ for $y > 0$.
- 5.8 A Pareto variable has $m = 1, \alpha = 1.5$. Does it have a finite mean? A finite variance? Can the usual finite-variance CLT be invoked?

5.9 Solutions and reasoning

- 5.1 $\text{Var}(X) = 3.2 - 1.4^2 = 1.24$; $\text{Var}(3X - 2) = 9(1.24) = 11.16$.
- 5.2 $\int_0^1 cx^2 dx = c/3 = 1$, hence $c = 3$. The CDF is x^3 inside $(0, 1)$, zero below and one above. Tail probability $1 - .8^3 = .488$.
- 5.3 Conditional on exceeding 10, the remaining location is uniform in $(10, 15)$. Exceeding 13 occupies length 2 of the remaining length 5, giving .4. The unconditional chance of exceeding 3 was .8, showing non-memorylessness.
- 5.4 Rate is $1/8$. By memorylessness the conditional probability is $\mathbb{P}(X > 7) = e^{-7/8}$. The median solves $1 - e^{-m/8} = 1/2$, so $m = 8 \log 2 \approx 5.545$ hours.
- 5.5 Standardized endpoints are $-1, 2$, giving $\Phi(2) - \Phi(-1) \approx .8186$. The percentile is $50 + 4(1.2816) = 55.1264$.
- 5.6 For $y \geq 0$, $\mathbb{P}(-\log X \leq y) = \mathbb{P}(X \geq e^{-y}) = 1 - e^{-y}$. Below zero the CDF is zero. Thus $Y \sim \text{Exp}(1)$.
- 5.7 $F_Y(y) = \Phi(\sqrt{y}) - \Phi(-\sqrt{y})$ for $y \geq 0$. Differentiating gives $f_Y(y) = \phi(\sqrt{y})/\sqrt{y} = e^{-y/2}/\sqrt{2\pi y}$. This is chi-square with one degree of freedom.
- 5.8 The mean is $1.5/.5 = 3$. The second moment and variance are infinite. The ordinary finite-variance CLT does not apply; the finite-mean LLN still does under iid sampling.

Class 6: averages, the LLN, and the CLT

6.1 Why this part exists

You measure an algorithm's runtime once. It takes 12 ms. That might be typical, or it might be an unusually quiet moment. Now measure it 100 times and average. Why should that help? How much should it help? How likely is the average to miss the true mean by more than 1 ms?

These are three different questions. Expectation defines the target. The law of large numbers explains stabilization. The central limit theorem gives an approximate shape and scale for the remaining error.

6.2 Continuous expectation and quantiles

For an integrable continuous X ,

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} xf(x) dx, \quad \mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x)f(x) dx.$$

The intuition is a fine partition: an interval near x has probability approximately $f(x)\Delta x$, contributing $xf(x)\Delta x$ to the weighted average. Passing to an integral gives the formula. More formally this is integration against the probability measure with density f .

Variance remains $\mathbb{E}[X^2] - \mu^2$. For $X \sim \text{Unif}(a, b)$,

$$\mathbb{E}[X] = \frac{1}{b-a} \left[\frac{x^2}{2} \right]_a^b = \frac{a+b}{2}, \quad \mathbb{E}[X^2] = \frac{a^2 + ab + b^2}{3},$$

so $\text{Var}(X) = (b-a)^2/12$.

For $X \sim \text{Exp}(\lambda)$, integration by parts gives

$$\mathbb{E}[X] = \int_0^{\infty} x\lambda e^{-\lambda x} dx = \frac{1}{\lambda}, \quad \mathbb{E}[X^2] = \frac{2}{\lambda^2}, \quad \text{Var}(X) = \frac{1}{\lambda^2}.$$

In the first integral choose $u = x$, $dv = \lambda e^{-\lambda x} dx$, so the boundary term is zero and the remaining integral is $\int_0^{\infty} e^{-\lambda x} dx$. Repeat once more for the second moment.

For $Z \sim \mathcal{N}(0, 1)$, symmetry gives $\mathbb{E}[Z] = 0$. Since $\phi'(z) = -z\phi(z)$, integration by parts gives $\int z^2\phi(z)dz = 1$ (the boundary $z\phi(z)$ vanishes). Therefore $X = \mu + \sigma Z$ has mean μ and variance σ^2 .

The q quantile is $F^{-1}(q) = \inf\{x : F(x) \geq q\}$. A median is a value with at least half the mass on each side (allowing equality at atoms). Quantiles answer service-level questions such as “what duration do 95% of jobs finish within?” Means answer average-cost questions. They are not interchangeable.

For exponential X , $q_{.95} = -\log(.05)/\lambda \approx 2.996/\lambda$, almost three times the mean. That makes a practical point: provisioning at the average is not the same as handling 95% of individual cases.

6.3 Three levels of randomness

1. The distribution of a single observation X , such as one runtime.
2. The distribution of a statistic, such as $\bar{X}_{100}$, across repeated batches of 100 observations.
3. One observed batch and its numerical mean $\bar{x}$.

The sample mean's distribution is called a *sampling distribution*. You need not literally collect thousands of batches to define it. It describes what would change if the data collection were repeated under the same model.

Theorem 6.1 (Mean and variance of an iid average). *If $X_1, \dots, X_n$ are iid with mean μ and variance $\sigma^2 < \infty$, then*

$$\mathbb{E}[\bar{X}_n] = \mu, \quad \text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}, \quad \text{SD}(\bar{X}_n) = \frac{\sigma}{\sqrt{n}}.$$

Proof. Linearity gives $n^{-1} \sum_i \mu = \mu$. Independence gives $\text{Var}(\sum_i X_i) = n\sigma^2$. Scaling by $1/n$ divides variance by n^2 , leaving σ^2/n . $\square$

The last quantity is the standard error of the mean. Individual observations do not become less variable when you collect more of them. Their average becomes less variable.

6.4 A rigorous route to the law of large numbers

Theorem 6.2 (Markov's inequality). *If $Y \geq 0$ and $a > 0$, $\mathbb{P}(Y \geq a) \leq \mathbb{E}[Y]/a$.*

Proof. Pointwise $Y \geq a\mathbf{1}_{\{Y \geq a\}}$. Taking expectations gives $\mathbb{E}[Y] \geq a\mathbb{P}(Y \geq a)$. $\square$

Apply Markov to $Y = (X - \mu)^2$ to obtain Chebyshev's inequality:

$$\mathbb{P}(|X - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}.$$

This bound does not assume a bell-shaped distribution.

Theorem 6.3 (Weak law, finite-variance version). *For iid variables with mean μ and finite variance, for every $\varepsilon > 0$,*

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \rightarrow 0.$$

Proof. Chebyshev and the variance-of-the-average formula give

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \rightarrow 0.$$

$\square$

The weak law says the chance of a fixed-size error vanishes. It does not say each new average is closer than the previous one. It does not say a streak must be compensated by the next trial. And it does not give a finite n at which the answer becomes exact.

The iid law in fact needs only $\mathbb{E}|X| < \infty$, not finite variance. Proving that extension uses truncation. The *strong* law states $\mathbb{P}(\lim_n \bar{X}_n = \mu) = 1$ under the same integrability condition. This is a statement about almost every infinite sample path, stronger than convergence in probability. The elementary proof above establishes the finite-variance weak law, not the full strong law.

Worked example: turning a desired error into a rigorous sample size

For Bernoulli observations, variance is at most $1/4$. To ensure $\mathbb{P}(|\hat{p} - p| \geq .02) \leq .05$, Chebyshev suffices if

$$\frac{1}{4n(.02)^2} \leq .05 \iff n \geq 12500.$$

This is a conservative finite-sample guarantee under independent sampling. Later a normal approximation will suggest about 2401 observations for a similar nominal target. Those answers differ because a general bound and an approximation are not the same claim.

6.5 The central limit theorem: a distribution for the error

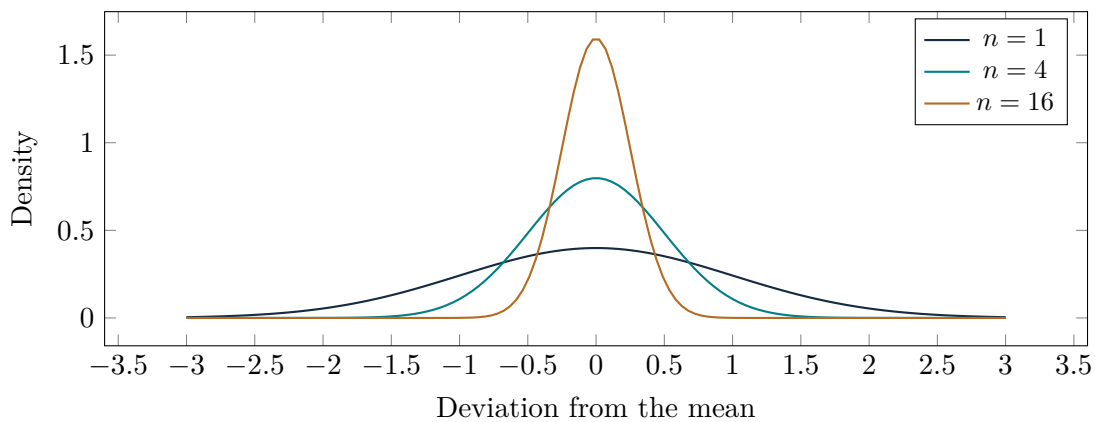
Theorem 6.4 (Classical iid CLT). *If X_i are iid with mean μ and $0 < \sigma^2 < \infty$, then*

$$Z_n = \frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}} = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1).$$

Equivalently, for every real z , $\mathbb{P}(Z_n \leq z) \rightarrow \Phi(z)$.

The notation $\xrightarrow{d}$ means convergence in distribution. The observations themselves do not become normal. The standardized sum or average has a distribution approaching normal.

Why standardize? Without zooming in, the average's distribution collapses around μ . Subtracting μ centers the error. Dividing by $\sigma/\sqrt{n}$ zooms in by exactly the amount needed to keep its variance equal to one. The CLT describes the shape visible under that changing magnification.



This exact normal-data example shows the average narrowing. Its unscaled density gets taller because the total area stays one. After standardization, all three curves would be the same standard normal curve. For non-normal finite-variance data, the standardized shape approaches normal rather than already being normal.

6.6 A proof of the CLT, with the advanced ingredient exposed

This subsection is an extension beyond the usual computational treatment. It proves the iid finite-variance CLT using characteristic functions; one standard theorem from advanced probability is stated, not proved from scratch.

For a real random variable Y , its characteristic function is $\varphi_Y(t) = \mathbb{E}[e^{itY}]$, where $i^2 = -1$. It always exists because $|e^{itY}| = 1$. Independence implies $\varphi_{Y+W}(t) = \varphi_Y(t)\varphi_W(t)$ by factoring the expectation.

Put $Y = (X - \mu)/\sigma$, so $\mathbb{E}Y = 0$ and $\mathbb{E}Y^2 = 1$. Taylor expansion under a finite-second-moment domination argument gives

$$\varphi_Y(u) = 1 - \frac{u^2}{2} + o(u^2), \quad u \rightarrow 0.$$

Here is the justification for the remainder rather than an unjustified formal expansion. Write $e^{iv} = 1 + iv - v^2/2 + r(v)$, where $r(v)/v^2 \rightarrow 0$ and $|r(v)| \leq Cv^2$ for a fixed constant C . For small v this follows from Taylor's theorem; for $|v| \geq 1$ it follows by bounding the other terms by a constant times v^2 . Then $r(uY)/u^2 \rightarrow 0$ pointwise and is bounded by CY^2 , which is integrable. Dominated convergence gives an $o(u^2)$ expected remainder.

For independent copies Y_i ,

$$\varphi_{Z_n}(t) = \left[\varphi_Y\left(\frac{t}{\sqrt{n}}\right) \right]^n = \left[1 - \frac{t^2}{2n} + o(n^{-1}) \right]^n \rightarrow e^{-t^2/2}.$$

The last limit follows from $\log(1+w) = w + o(w)$ near zero. The characteristic function of a standard normal is $e^{-t^2/2}$: differentiate its defining integral and integrate by parts to obtain $\varphi'(t) = -t\varphi(t)$ with $\varphi(0) = 1$.

Imported theorem (Lévy's continuity theorem). If characteristic functions converge pointwise to a characteristic function continuous at zero, their distributions converge to that limiting distribution. Applying it proves the CLT. The proof of this continuity theorem and the full integration framework belong to a more advanced probability course; they are not being disguised as a one-line elementary fact.

6.7 Using the CLT correctly

For large enough n under the assumptions,

$$\sum_i X_i \approx \mathcal{N}(n\mu, n\sigma^2), \quad \bar{X}_n \approx \mathcal{N}(\mu, \sigma^2/n).$$

First identify whether the problem asks about a sum, an average, or one observation. Then calculate the correct center and SD. Only then standardize.

Worked example: an average runtime threshold

Suppose iid job times have mean 20 ms and SD 8 ms. For a batch of 100 jobs,

$$\mathbb{P}(\bar{X}_{100} > 22) \approx 1 - \Phi\left(\frac{22 - 20}{8/10}\right) = 1 - \Phi(2.5) \approx .0062.$$

This answers whether the *batch average* exceeds 22, not whether an individual job exceeds 22. The latter cannot be determined from the mean and variance alone without a distribution model.

Worked example: a capacity problem about a sum

Suppose 400 independent package weights have mean 2 kg and SD .5 kg. Their total has mean 800 and SD $\sqrt{400}(.5) = 10$. Therefore $\mathbb{P}(\text{total} > 820) \approx 1 - \Phi(2) = .0228$. Using SD $400(.5) = 200$ would incorrectly assume their centered fluctuations move together perfectly.

Worked example: binomial approximation and continuity correction

Let $K \sim \text{Bin}(200, .4)$. Then $\mathbb{E}K = 80$ and $\text{SD}(K) = \sqrt{48}$. To approximate $\mathbb{P}(K \geq 90)$, cover the integer bars starting at 90 with a continuous area starting at 89.5:

$$\mathbb{P}(K \geq 90) \approx 1 - \Phi\left(\frac{89.5 - 80}{\sqrt{48}}\right) \approx .0852.$$

The half-unit continuity correction often improves a discrete-to-continuous approximation. Exact binomial evaluation remains preferable when convenient. Rules like np and $n(1 - p)$ both being reasonably large are heuristics, not a universal error guarantee.

6.8 Why “ $n \geq 30$ ” is not a theorem

Highly skewed distributions may converge slowly; rare extreme events can dominate. A heavy tail with infinite variance invalidates this CLT. Strong dependence may prevent standard-error shrinkage entirely. A systematic measurement bias does not average away.

If $X_i = Y$ for all i , then $\bar{X}_n = Y$ no matter how large n is. If a scale adds a fixed 2 kg bias, repeated measurements average toward the biased mean, not the correct weight. More data cannot repair every problem.

6.9 Histograms and the LLN behind them

For bins B_j of width h_j , the density-histogram height is

$$\hat{f}_j = \frac{\text{number of observations in bin } j}{nh_j}.$$

Height times width equals the observed fraction, so total bar area is one. With unequal bin widths, raw counts are not density heights.

For a fixed bin, $I_{ij} = \mathbf{1}_{\{X_i \in B_j\}}$ is Bernoulli with mean $p_j = \mathbb{P}(X \in B_j)$. The LLN gives its observed fraction converging to p_j . For finitely many fixed bins, a union bound proves all the fractions converge simultaneously in probability.

To connect that with a smooth PDF, assume f is continuous on a compact interval. Uniform continuity lets us choose sufficiently narrow fixed bins so that each bin-average density p_j/h_j is uniformly close to f inside the bin. Then choose n large enough that all estimated bin heights are close to their bin averages. This gives an elementary two-stage argument. Shrinking bins with n requires additional conditions; merely making bins arbitrarily tiny on a fixed data set produces noise, not precision.

6.10 Practice: try before reading the solutions

- 6.1 $X \sim \text{Unif}(2, 8)$. Find its mean, variance, and 75th percentile.
- 6.2 An iid population has mean 12 and SD 6. Find the mean and SD of a sum of 64 observations and of their average.
- 6.3 With the same population, give a Chebyshev bound for $\mathbb{P}(|\bar{X}_{100} - 12| \geq 2)$.
- 6.4 iid waiting times have exponential rate 2 per minute. Approximate the probability that the total of 100 waits exceeds 60 minutes.

- 6.5 For $K \sim \text{Bin}(100, .5)$, approximate $\mathbb{P}(45 \leq K \leq 55)$ using continuity correction.
- 6.6 A poll uses iid Bernoulli data. Use Chebyshev to guarantee absolute error below .05 with probability at least .99. How many observations suffice?
- 6.7 A histogram bin of width 4 contains 30 of 200 observations. What is its density height? What does its area mean?
- 6.8 Explain what is wrong with: “The CLT says that after observing many exponential values, the exponential distribution becomes normal.”
- 6.9 Show that for independent mean-zero errors of variance 9, averaging 100 errors has SD .3, whereas summing them has SD 30.
- 6.10 A program generates one random bit and repeats it one million times. What is its average? Why does the iid LLN not justify an average close to .5?

6.11 Solutions and reasoning

- 6.1 Mean 5, variance $6^2/12 = 3$, quantile $2 + .75(6) = 6.5$.
- 6.2 Sum: mean 768, SD 48. Average: mean 12, SD .75. The scaling distinction is the entire point of this exercise.
- 6.3 Variance of the average is $36/100 = .36$. Divide by 2^2 to get .09. This is an upper bound, not an approximate equality.
- 6.4 One wait has mean .5 and variance .25. The total has mean 50 and SD 5; CLT gives $1 - \Phi(2) \approx .0228$. The exact sum is gamma distributed; no normality of individual waits is asserted.
- 6.5 Mean 50 and SD 5. Use continuous interval $[44.5, 55.5]$, giving $\Phi(1.1) - \Phi(-1.1) \approx .7287$.
- 6.6 Require $1/[4n(.05)^2] \leq .01$, hence $n \geq 10000$.
- 6.7 Height is $30/(200 \cdot 4) = .0375$ per horizontal unit. Its area .15 is the observed fraction of data in the bin.
- 6.8 Each observation retains its exponential law. The standardized mean of a large independent batch is approximately normal across repeated batches.
- 6.9 A sum has variance $100 \cdot 9 = 900$, hence SD 30. Divide by 100 to average, giving SD .3. Equivalently average variance is $9/100$.
- 6.10 The average equals the original bit, so it is zero or one. All entries are perfectly dependent; increasing their count does not create independent information.

Class 7: joint behavior, covariance, and correlation

7.1 Why this part exists

Suppose frontend and backend response times each average 50 ms. To understand total page latency, you must know whether slow frontend events tend to coincide with slow backend events. Separate distributions cannot answer that.

A joint distribution records how two variables behave together. Marginals discard the relationship. Covariance summarizes one important aspect of it, but not all of it.

7.2 Joint PMFs and marginals

For discrete X, Y , define $p_{X,Y}(x, y) = \mathbb{P}(X = x, Y = y)$. It is nonnegative and sums to one over all pairs. Marginalization sums out the unwanted coordinate:

$$p_X(x) = \sum_y p_{X,Y}(x, y), \quad p_Y(y) = \sum_x p_{X,Y}(x, y).$$

Consider a joint table:

	$Y = 0$	$Y = 1$	p_X
$X = 0$	.4	.1	.5
$X = 1$	.2	.3	.5
p_Y	.6	.4	1

Here $\mathbb{P}(X = 1 \mid Y = 1) = .3/.4 = .75$, not .5. The joint structure matters even though the marginal distribution of X is a fair Bernoulli.

The joint CDF is $F(x, y) = \mathbb{P}(X \leq x, Y \leq y)$. Rectangle probabilities use two-dimensional inclusion-exclusion:

$$\begin{aligned} \mathbb{P}(a < X \leq b, c < Y \leq d) &= F(b, d) - F(a, d) \\ &\quad - F(b, c) + F(a, c). \end{aligned}$$

7.3 Joint densities and integration regions

For a joint density $f(x, y)$,

$$\mathbb{P}((X, Y) \in A) = \iint_A f(x, y) dx dy, \quad f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy.$$

The support is part of the model. Draw or describe it before integrating. A formula that looks like a product can still describe dependent variables if the support constrains the coordinates together.

Worked example: a uniform point in a triangle

Let $f(x, y) = 2$ on $0 < y < x < 1$ and zero elsewhere. The triangle has area $1/2$, so the density integrates to one. Marginalizing yields

$$f_X(x) = \int_0^x 2 dy = 2x, \quad f_Y(y) = \int_y^1 2 dx = 2(1 - y), \quad 0 < x, y < 1.$$

The event $Y < X/2$ occupies half the vertical length at every x , so

$$\mathbb{P}(Y < X/2) = \int_0^1 \int_0^{x/2} 2 dy dx = \frac{1}{2}.$$

But X, Y are not independent. For example, Y cannot exceed X , although each separately can take any value between zero and one.

For $f_X(x) > 0$, the conditional density is $f_{Y|X=x}(y) = f(x, y)/f_X(x)$. In this example it is $1/x$ on $(0, x)$, so given $X = x$, Y is uniform there. This is a density-based conditional definition, not the elementary ratio of two zero-probability events.

7.4 Independence of random variables

X and Y are independent when $\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B)$ for all appropriate sets A, B . Equivalently, their joint CDF factors. For discrete variables, this is equivalent to $p(x, y) = p_X(x)p_Y(y)$ everywhere. With densities, factorization holds almost everywhere.

Under independence, integrable products factor:

$$\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)]$$

whenever the expectations are well defined with suitable integrability. Proof: substitute the product density or PMF into the double integral or sum and separate the factors.

7.5 Covariance measures signed co-movement

For finite second moments,

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}[XY] - \mu_X\mu_Y.$$

Both above average or both below average gives a positive product. Opposite directions gives a negative product. Covariance averages these signed products. It has product units, so its numerical size changes with the measurement scales.

From the definition, covariance is symmetric and bilinear; constants have zero covariance. In particular,

$$\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y).$$

Expanding the squared centered sum yields

$$\boxed{\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y)}.$$

For a collection, $\text{Var}(\sum_i X_i) = \sum_i \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j)$. This formula is the reason shared shocks prevent averaging from working as well as independent-noise calculations predict.

Worked example: calculating the joint table's covariance

In the earlier table, $\mathbb{E}X = .5$, $\mathbb{E}Y = .4$, and $\mathbb{E}[XY] = .3$ because the product is nonzero only at $(1, 1)$. Thus $\text{Cov}(X, Y) = .3 - .2 = .1$. Their variances are .25 and .24, giving correlation about .4082.

Worked example: the triangle's moments

Using its marginals, $\mathbb{E}X = \int_0^1 2x^2 dx = 2/3$ and $\mathbb{E}Y = \int_0^1 2y(1-y) dy = 1/3$. Also

$$\mathbb{E}[XY] = \int_0^1 \int_0^x 2xy dy dx = \int_0^1 x^3 dx = \frac{1}{4}.$$

Hence covariance is $1/4 - 2/9 = 1/36$. Both variances equal $1/18$, giving correlation $1/2$. The positive relationship is unsurprising: a larger X allows Y to occupy a larger interval.

7.6 Correlation removes the units

When $\sigma_X, \sigma_Y > 0$, define

$$\rho = \text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

It is the expected product of the standardized variables. It measures linear association, not general dependence and not causation.

Theorem 7.1. $-1 \leq \rho \leq 1$. Equality in absolute value occurs exactly when the two variables have an almost-sure affine relationship with nonzero slope.

Proof. Let $U = (X - \mu_X)/\sigma_X$ and $V = (Y - \mu_Y)/\sigma_Y$. Then $\mathbb{E}[(U - V)^2] = 2 - 2\rho \geq 0$ and $\mathbb{E}[(U + V)^2] = 2 + 2\rho \geq 0$. Thus the bounds hold. Equality $\rho = 1$ forces the nonnegative variable $(U - V)^2$ to have mean zero, hence $U = V$ almost surely. Similarly $\rho = -1$ forces $U = -V$. Rearranging gives the affine relation, and the converse follows immediately. $\square$

If one variable is constant its SD is zero and correlation is undefined, not zero.

Independence implies zero covariance because $\mathbb{E}[XY] = \mathbb{E}X\mathbb{E}Y$. The converse is false. Let $X \sim \text{Unif}(-1, 1)$ and $Y = X^2$. Symmetry gives $\mathbb{E}X = \mathbb{E}X^3 = 0$, so covariance is zero. Yet Y is determined exactly by X : the dependence is quadratic, not linear.

Worked example: averages with a common shock

Suppose $X_i = U + \varepsilon_i$, where a shared shock U is independent of iid individual errors ε_i , all with mean zero. Then

$$\bar{X}_n = U + \bar{\varepsilon}_n, \quad \text{Var}(\bar{X}_n) = \text{Var}(U) + \frac{\text{Var}(\varepsilon_1)}{n}.$$

More observations reduce the individual noise but cannot remove the common shock. This models a benchmark batch affected by the same background load, or many measurements made with the same calibration error.

7.7 Useful extension: conditioning to split variability

The law of total expectation says $\mathbb{E}X = \mathbb{E}[\mathbb{E}(X | Y)]$. For discrete variables, substitute the conditional PMF and sum: $\sum_y p_Y(y) \sum_x xp(x | y) = \sum_{x,y} xp(x, y) = \mathbb{E}X$. Integrals give the continuous version.

The law of total variance is

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X | Y)] + \text{Var}(\mathbb{E}[X | Y]).$$

Proof: write $X - \mathbb{E}X = (X - \mathbb{E}[X | Y]) + (\mathbb{E}[X | Y] - \mathbb{E}X)$, square, and condition on Y . The cross term vanishes since $\mathbb{E}[X - \mathbb{E}(X | Y) | Y] = 0$. The formula separates within-group noise from between-group differences. It is often the right tool when observations come from several machines or populations.

7.8 Practice: try before reading the solutions

- 7.1 In the joint table, find $\mathbb{P}(X = 1 \text{ or } Y = 1)$ and $\mathbb{P}(Y = 1 | X = 0)$.
- 7.2 Let $f(x, y) = 4xy$ on the unit square. Check normalization, find marginals, and decide independence.
- 7.3 For independent X, Y with variances 4 and 9, find $\text{Var}(2X - 3Y)$. What changes if covariance is 1?
- 7.4 If $Y = 7 - 2X$ and $0 < \text{Var}(X) < \infty$, find the correlation.
- 7.5 Two job times each have variance 100 and correlation .6. Find the variance and SD of their sum.
- 7.6 For the triangular model, find $\mathbb{P}(X > .8)$ and $\mathbb{E}[Y | X = x]$.
- 7.7 Give a dependent pair with zero covariance, and explain why zero covariance cannot prove independence.
- 7.8 In the common-shock model, $\text{Var}(U) = 4$ and $\text{Var}(\varepsilon_i) = 9$. Find the SD of the average for $n = 100$ and its limiting SD as $n \rightarrow \infty$.

7.9 Solutions and reasoning

- 7.1 The union is the complement of $(0, 0)$, giving .6. The conditional probability is $.1/.5 = .2$.
- 7.2 $\int_0^1 \int_0^1 4xy \, dy \, dx = 1$. Marginals are $2x$ and $2y$, whose product is exactly the joint density on a product support. Therefore independence holds.
- 7.3 Independent case: $4(4) + 9(9) = 97$. With covariance 1, add $2(2)(-3)(1) = -12$, giving 85. The sign of the coefficients matters.
- 7.4 Covariance is $-2 \text{Var}(X)$, while $\sigma_Y = 2\sigma_X$, so correlation is -1 .
- 7.5 Covariance is $.6(10)(10) = 60$. Sum variance is $100 + 100 + 120 = 320$, SD $\sqrt{320} \approx 17.889$.
- 7.6 $\mathbb{P}(X > .8) = \int_{.8}^1 2x \, dx = .36$. The conditional uniform distribution of Y on $(0, x)$ has mean $x/2$.
- 7.7 Use $X \sim \text{Unif}(-1, 1)$, $Y = X^2$. Covariance averages a signed linear product and misses this symmetric nonlinear dependence. Knowing X nevertheless determines Y .
- 7.8 Variance is $4 + 9/100 = 4.09$, SD about 2.0224. The limiting SD is 2, not zero, because the shared shock remains.

Class 8: putting probability together

8.1 Why this part exists

In a mixed problem nobody labels the required method. Recognizing the mathematical structure is part of the skill. This chapter replaces the official review session with a compact diagnostic toolkit and integrated problems.

8.2 Start with the question, not the formula

What the problem asks	Likely starting point
How many possible selections?	Counting; distinguish order and replacement.
At least one event occurs?	Complement or union bound; exact inclusion-exclusion if useful.
Given some information?	Restrict and renormalize; write the conditional explicitly.
Cause given evidence?	Bayes plus total probability.
Number of successes in n trials?	Binomial if independent with common p .
Trials until the first success?	Geometric, with its support convention stated.
Continuous duration or measurement?	Specify a density or a mechanism; integrate.
A transformed observation?	CDF method; check all inverse branches.
Average cost or count?	Expectation; indicators and linearity.
Variability of a sum?	Variances plus covariance terms.
Large-sample sum or average tail?	CLT after checking assumptions and scaling.
Distribution-free tail bound?	Markov or Chebyshev.
Two quantities together?	Joint distribution, support, marginalization.

Five lines that prevent most mistakes

Write the random variable in words; state its support; name or define the model; list the assumptions; translate the requested event into symbols. A formula should appear only after these lines.

8.3 Integrated example: a two-stage request

A request goes through authentication and then data retrieval. Authentication fails with probability .02. Conditional on authentication succeeding, retrieval fails with probability .03. The probability of complete success is $.98(.97) = .9506$. We used a conditional probability, not an unstated independence assumption.

If separate requests are iid under this model, the failure count among 500 requests is binomial with $p = .0494$, mean 24.7 and variance $500(.0494)(.9506)$. If every failure is independently retried until success under the same conditions, expected attempts per completed operation is $1/.9506$. If failures trigger a shared outage, these iid and geometric assumptions become questionable.

Suppose a failure is observed. Authentication failure is one possible route; retrieval failure after successful authentication is the other. Therefore

$$\mathbb{P}(\text{authentication failed} \mid \text{request failed}) = \frac{.02}{.0494} \approx .4049.$$

One scenario uses multiplication, total probability, Bayes, binomial counts, and geometric waiting. The formulas are connected because they describe different questions about the same model.

8.4 Integrated example: choose exact, approximate, or bounded

Let $K \sim \text{Bin}(100, .1)$. To find $\mathbb{P}(K \geq 15)$ you have three kinds of answer:

$$\text{Exact: } \sum_{k=15}^{100} \binom{100}{k} (.1)^k (.9)^{100-k}.$$

The CLT uses mean 10 and SD 3, giving approximately $1 - \Phi((14.5 - 10)/3) = 1 - \Phi(1.5) \approx .0668$. Chebyshev bounds $\mathbb{P}(|K - 10| \geq 5)$ by $9/25 = .36$, hence also bounds the desired one-sided event by .36.

The exact expression is correct by definition; the approximation is convenient but imperfect; the bound is rigorous but loose. Never report all three as though they were interchangeable estimates.

8.5 Practice: try before reading the solutions

- 8.1 A random sample of five devices is taken without replacement from 30 devices, of which six are faulty. Find the probability of exactly two faulty devices and the expected number of faulty devices.
- 8.2 Independent attempts succeed with probability .3 and take 2 seconds each. Find expected completion time and probability of completion within 6 seconds.
- 8.3 $X \sim \text{Exp}(1)$ and $Y = 2X + 3$. Find the support, CDF, mean, and variance of Y .
- 8.4 Independent task times have mean 5 and SD 2. Approximate the chance that 100 tasks take more than 540 total time units.
- 8.5 X, Y are independent uniform variables on $(0, 1)$. Find $\mathbb{P}(X + Y < 1/2)$ geometrically.
- 8.6 A random variable has mean 7 and variance 4. Is $\mathbb{P}(X > 11)$ determined? Give a valid upper bound.

8.6 Solutions and reasoning

- 8.1 The probability is $\binom{6}{2}\binom{24}{3}/\binom{30}{5}$. The count is hypergeometric, not binomial. Each sampled position has marginal faulty probability $6/30$, so linearity gives expected count $5(6/30) = 1$, despite dependence.
- 8.2 Trial count is geometric with mean $1/.3$, so expected time is $2/.3 = 20/3$ seconds. At most three attempts fit in six seconds; probability of success by then is $1 - (.7)^3 = .657$.

- 8.3 Support is $[3, \infty)$. For $y \geq 3$, $F_Y(y) = 1 - e^{-(y-3)/2}$; below 3 it is zero. Mean is $2(1) + 3 = 5$, variance $2^2(1) = 4$.
- 8.4 Total mean 500 and SD $2\sqrt{100} = 20$. Probability is approximately $1 - \Phi(2) = .0228$. If the observations are strongly dependent the calculation needs modification.
- 8.5 The unit-square density is one. The desired region is a right triangle with legs $1/2$, area $1/8$.
- 8.6 Mean and variance do not determine a unique tail probability. Chebyshev gives $\mathbb{P}(X > 11) \leq \mathbb{P}(|X - 7| \geq 4) \leq 4/16 = 1/4$.

Chapter 9

Class 9: original probability practice exam

The corresponding MIT session is Exam 1. This is an original 100-point practice paper, not a reproduction of an MIT exam. Allow 100 minutes. Show the model and reasoning. You may use $\Phi(1) = .8413$, $\Phi(1.5) = .9332$, $\Phi(2) = .9772$.

9.1 Questions

- 9.1 **Counting and sampling, 12 points.** A committee of four is selected uniformly from seven programmers and five designers. (a) Find the probability of exactly two designers. (b) Find the probability of at least one designer. (c) Explain why a binomial calculation is not exact.
- 9.2 **Bayes, 14 points.** A machine chooses source A with probability .6, source B with probability .4. Conditional on the source, a produced item is defective with probability .02 for A and .08 for B. (a) Find the defect probability. (b) An item is defective; find the posterior probability of B. (c) Give one assumption needed to multiply defect probabilities for two items.
- 9.3 **Expectation without a full distribution, 10 points.** Ten balls independently choose one of five boxes uniformly. Let N be the number of empty boxes. Find $\mathbb{E}[N]$. Explain why the box-empty indicators need not be independent for your argument.
- 9.4 **A continuous model, 16 points.** Let $f(x) = cx$ on $(0, 2)$. Find c , its CDF, mean, variance, and median.
- 9.5 **Transformations, 12 points.** Let $X \sim \text{Unif}(-2, 2)$ and $Y = X^2$. Find the support and CDF of Y , its density, and $\mathbb{E}[Y]$ using the original variable X .
- 9.6 **Joint distribution, 16 points.** Let $f(x, y) = 2$ on $0 < y < x < 1$. Find both marginals, $\mathbb{E}[XY]$, covariance, and whether the variables are independent.
- 9.7 **LLN and CLT, 14 points.** iid variables have mean 30 and SD 10. For $n = 100$, (a) approximate $\mathbb{P}(\bar{X} > 32)$; (b) bound $\mathbb{P}(|\bar{X} - 30| \geq 2)$ using Chebyshev; (c) explain why the two answers need not agree.
- 9.8 **Conceptual check, 6 points.** Explain why zero correlation does not imply independence, why a density can exceed one, and why collecting more observations does not make individual observations less variable.

9.2 Worked solutions

- 9.1 (a) Favorable committees contain two designers and two programmers, so the answer is $\binom{5}{2}\binom{7}{2}/\binom{12}{4} = 210/495 = 14/33$. (b) Exclude all-programmer committees: $1 - \binom{7}{4}/\binom{12}{4} = 1 - 35/495 = 92/99$. (c) Sampling without replacement makes successive categories dependent and changes conditional success probabilities. The appropriate count distribution is hypergeometric.
- 9.2 (a) Total probability gives $.6(.02) + .4(.08) = .044$. (b) The B-and-defect probability is $.032$, so posterior $.032/.044 = 8/11$. (c) Independence must be justified at the relevant level: if the same source is shared, condition on it before factoring, and then average over sources. If sources are chosen independently for each item, unconditional independence can hold under additional independent-production assumptions.
- 9.3 Let I_j indicate box j is empty. Each ball avoids it with probability $4/5$, so $\mathbb{E}[I_j] = (4/5)^{10}$. Sum over five boxes: $\mathbb{E}[N] = 5(4/5)^{10} \approx .5369$. Linearity does not require the I_j to be independent. Indeed, occupied capacity creates relationships among boxes.
- 9.4 Normalize: $2c = 1$, hence $c = 1/2$. CDF is zero below 0, $x^2/4$ in $(0, 2)$, and one above 2. Mean $\int_0^2 x(x/2)dx = 4/3$; second moment $\int_0^2 x^2(x/2)dx = 2$; variance $2 - 16/9 = 2/9$. Median solves $m^2/4 = 1/2$, giving $m = \sqrt{2}$. The mean and median differ because the density is not symmetric.
- 9.5 Support is $[0, 4]$. For $0 \leq y \leq 4$, the allowed X interval is $[-\sqrt{y}, \sqrt{y}]$, length $2\sqrt{y}$ out of total length 4. Thus $F_Y(y) = \sqrt{y}/2$, with exterior values zero and one. Density is $1/(4\sqrt{y})$ for $0 < y < 4$. Finally $\mathbb{E}[Y] = \mathbb{E}[X^2] = (1/4) \int_{-2}^2 x^2 dx = 4/3$.
- 9.6 Marginals are $2x$ and $2(1-y)$ on $(0, 1)$. The joint product moment is $\int_0^1 \int_0^x 2xy \, dy \, dx = 1/4$. Means are $2/3$ and $1/3$, so covariance $1/36$. Factorization fails: the joint density is zero above the diagonal while the product of marginals is positive there. Hence they are dependent.
- 9.7 (a) Average SD is $10/\sqrt{100} = 1$, giving tail approximately $1 - \Phi(2) = .0228$. (b) Average variance is 1, so the two-sided error probability is bounded by $1/4$. (c) The first concerns a one-sided tail and is approximate; the second concerns two tails and is a rigorous upper bound for any distribution satisfying the stated moment and independence assumptions. Even the CLT two-sided approximation $.0456$ can be much smaller than the upper bound $.25$.
- 9.8 The example $Y = X^2$ with symmetric X gives dependence with zero covariance. A density measures probability per unit length, and only its integral is constrained to one. More samples leave the distribution of each X_i unchanged while changing the distribution of $\bar{X}_n$.

9.3 How to diagnose your mistakes

A wrong denominator often signals a sample-space mismatch. A missing $\sqrt{n}$ usually means confusing observations with averages. A correct integral over the wrong region is still the wrong answer. A correct numerical Bayes calculation with an unjustified independence assumption is an incomplete model. Redo each missed problem by writing the five-line setup from class 8.

Class 10: statistical inference and maximum likelihood

10.1 Why this part exists

Up to now a distribution's parameters have mostly been handed to you. A real engineer does not know the failure probability, the average runtime, or the event rate. We observe data and try to infer those unknown quantities.

Maximum likelihood asks: among the models under consideration, which parameter value makes the data we actually observed most compatible with the model? It is an optimization problem with a statistical interpretation.

10.2 Parameters, data, statistics, estimators

A parametric model is a family $\{f(x | \theta) : \theta \in \Theta\}$. The parameter θ could be one number, such as an exponential rate, or a vector, such as (μ, σ^2) for a normal model.

Before collecting data, $X_1, \dots, X_n$ are random. After collecting them, $x_1, \dots, x_n$ are fixed observations. A statistic is a function of the data and known constants, with no unknown parameters in its formula. An estimator is a statistic used to estimate a parameter. Its observed numerical value is an estimate.

Thus $\bar{X}$ is an estimator of μ , and $\bar{x} = 17.4$ is an estimate. But $\bar{X} - \mu$ is not a statistic when μ is unknown, even though it is a mathematically meaningful random error. This distinction is crucial for confidence intervals.

10.3 Likelihood: freeze the data, vary the parameter

For observed data $x = (x_1, \dots, x_n)$, the likelihood is the joint PMF or density, now viewed as a function of θ :

$$L(\theta; x) = f(x | \theta).$$

Under independent observations with a common parameter,

$$L(\theta; x) = \prod_{i=1}^n f(x_i | \theta), \quad \ell(\theta; x) = \log L(\theta; x) = \sum_{i=1}^n \log f(x_i | \theta).$$

If the observations are dependent, use their joint law; a product is not automatic.

Likelihood is not a posterior probability

$L(\theta; x)$ compares data compatibility across parameter values. It need not integrate to one as a function of θ . For continuous data it is a density, not the probability of observing those exact real numbers. A posterior requires a prior and Bayes normalization.

For a continuous observation, consider a small recording interval around each x_i . Its probability is approximately the density times the interval width. If the same recording widths are used for every candidate parameter, those factors do not affect which parameter maximizes the likelihood. The MLE is $\hat{\theta} \in \arg \max_{\theta \in \Theta} L(\theta; x)$. Because logarithm is strictly increasing, maximizing L and ℓ gives the same answer wherever $L > 0$. Logs turn products into sums and reduce numerical underflow.

10.4 Worked derivations of the main MLEs

Bernoulli and binomial probability

Observe k successes in n independent Bernoulli trials. Ignoring a multiplicative factor independent of p ,

$$L(p) = p^k(1-p)^{n-k}, \quad \ell(p) = k \log p + (n-k) \log(1-p).$$

For $0 < k < n$,

$$\ell'(p) = \frac{k}{p} - \frac{n-k}{1-p} = 0 \implies \hat{p} = \frac{k}{n}.$$

The second derivative is $-k/p^2 - (n-k)/(1-p)^2 < 0$, so it is the unique maximum. If $k = 0$ the maximum is at $p = 0$; if $k = n$ it is at $p = 1$. Boundary cases must not disappear when taking derivatives.

For example, 18 failures in 300 requests yields $\hat{p} = .06$. This is a point estimate, not a statement that the unknown probability equals .06 exactly. If you observed only the count rather than the full sequence, the binomial coefficient enters the likelihood but is independent of p , so the MLE is unchanged.

Exponential rate

For positive observations from an exponential model,

$$\ell(\lambda) = n \log \lambda - \lambda \sum_i x_i, \quad \ell'(\lambda) = n/\lambda - \sum_i x_i.$$

The second derivative is $-n/\lambda^2 < 0$, giving

$$\hat{\lambda} = \frac{n}{\sum_i x_i} = \frac{1}{\bar{x}}.$$

For waits .2, .4, .3, .1 seconds, $\bar{x} = .25$ and $\hat{\lambda} = 4$ per second. These observations must be complete, uncensored waits under a common rate. An observation recorded only as “longer than five seconds” contributes a survival probability, not an ordinary density at five.

Normal mean and variance

Write $v = \sigma^2 > 0$. For iid normal data,

$$\ell(\mu, v) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log v - \frac{1}{2v} \sum_i (x_i - \mu)^2.$$

For any fixed v , maximizing means minimizing the sum of squared residuals. The identity

$$\sum_i (x_i - \mu)^2 = \sum_i (x_i - \bar{x})^2 + n(\bar{x} - \mu)^2$$

shows that $\hat{\mu} = \bar{x}$ minimizes it. Differentiating in v at that mean gives

$$\hat{v}_{\text{MLE}} = \frac{1}{n} \sum_i (x_i - \bar{x})^2.$$

This differs from the unbiased sample variance, which divides by $n - 1$. Maximum likelihood does not promise unbiasedness. If all observations coincide, the unconstrained positive-variance likelihood is unbounded as $v \downarrow 0$; there is no ordinary positive-variance maximizer. Nondegenerate normal data avoid that case almost surely for $n \geq 2$.

Uniform endpoints: support restrictions matter

For iid uniform observations on unknown $[a, b]$ with $a < b$,

$$L(a, b) = (b - a)^{-n} \mathbf{1}_{\{a \leq x_{\min}, b \geq x_{\max}\}}.$$

For a nondegenerate sample, the shortest interval containing all data has endpoints $\hat{a} = x_{\min}$ and $\hat{b} = x_{\max}$. Differentiation alone misses this boundary solution. Under a $\text{Unif}(0, \theta)$ model, $\hat{\theta} = x_{\max}$. This estimate usually undershoots the true endpoint because a finite sample rarely reaches it.

10.5 Applied likelihood examples

Worked example: capture-recapture

Mark M animals, release them, then later capture K animals without replacement. Observe $r > 0$ marked ones. If the population has unknown integer size N , uniform capture and a stable population give the hypergeometric likelihood

$$L(N) = \frac{\binom{M}{r} \binom{N-M}{K-r}}{\binom{N}{K}}, \quad N \geq M + K - r.$$

With $M = 15, K = 20, r = 4$, a useful initial estimate is $MK/r = 75$. We can determine the integer maximum by comparing neighbors:

$$\frac{L(N+1)}{L(N)} = \frac{(N+1-M)(N+1-K)}{(N+1-M-K+r)(N+1)}.$$

This ratio exceeds one exactly when $MK > r(N+1)$. Here it exceeds one up to the neighborhood of 75 and equals one at $N = 74$, so both 74 and 75 maximize the likelihood. Integer MLEs can tie. If $r = 0$, the likelihood may increase toward a supremum as $N \rightarrow \infty$, with no finite MLE.

The assumptions matter: animals must mix, tags must remain, the population must stay effectively closed, and capture must not selectively favor tagged animals. The formula does not repair a broken sampling mechanism.

Worked example: estimating a frequency from paired categories

Suppose an idealized model forms pairs by independently drawing labels A or a, with A probability θ . The pair categories have probabilities θ^2 , $2\theta(1-\theta)$, and $(1-\theta)^2$. Observe counts n_{AA}, n_{Aa}, n_{aa} across n independent pairs. The likelihood, up to constants, is

$$L(\theta) \propto \theta^{2n_{AA}+n_{Aa}} (1-\theta)^{2n_{aa}+n_{Aa}}.$$

This is the Bernoulli optimization with $2n$ labels, so

$$\hat{\theta} = \frac{2n_{AA} + n_{Aa}}{2n}.$$

For counts 20, 50, 30, the answer is .45. This is the algebra behind the introductory Hardy-Weinberg likelihood example; actual genetic interpretation requires the model's biological assumptions.

10.6 What makes an estimator good?

An estimator is unbiased if $\mathbb{E}_\theta[\hat{\theta}] = \theta$ for every allowed θ . It is consistent if $\hat{\theta}_n \rightarrow \theta$ in probability as sample size grows. These are different properties.

The mean squared error decomposes as

$$\mathbb{E}_\theta[(\hat{\theta} - \theta)^2] = \text{Var}_\theta(\hat{\theta}) + (\mathbb{E}_\theta\hat{\theta} - \theta)^2.$$

Proof: add and subtract $\mathbb{E}\hat{\theta}$, expand the square, and note the cross term has mean zero. A slightly biased estimator can have smaller total MSE if it reduces variance enough. MLEs often have useful large-sample properties under regularity conditions, but none of uniqueness, existence, unbiasedness, or finite-sample optimality is automatic.

10.7 Practice: try before reading the solutions

- 10.1 In 80 iid Bernoulli trials, 12 succeed. Derive the MLE and state what happens if instead zero succeed.
- 10.2 Exponential observations are 1, 2, 3, 4 hours. Find the MLE of the rate and of the mean waiting time.
- 10.3 Normal observations are 2, 4, 6. Find the MLEs of mean and variance, and compare the unbiased sample variance.
- 10.4 A sample from $\text{Unif}(0, \theta)$ is 2, 5, 7. Find the MLE and explain why $\theta = 6$ is impossible under the model.
- 10.5 Prove that the Bernoulli sample proportion is unbiased and has MSE $p(1-p)/n$.
- 10.6 If $M_n = \max(X_1, \dots, X_n)$ for iid $\text{Unif}(0, \theta)$, derive its CDF and mean. Construct an unbiased estimator of θ .
- 10.7 Why is maximizing a normal likelihood equivalent to least squares when the variance is fixed?
- 10.8 One data point is recorded as $X > 5$ under an exponential model. Write its likelihood contribution and explain why treating it as $X = 5$ is wrong.

10.8 Solutions and reasoning

- 10.1 Set $12/p - 68/(1-p) = 0$, obtaining $\hat{p} = 12/80 = .15$. With zero successes, $(1-p)^{80}$ decreases on $[0, 1]$, so the maximum occurs at the boundary $p = 0$.
- 10.2 Total time 10, $n = 4$, hence $\hat{\lambda} = .4$ per hour. By the one-to-one transformation $\mu = 1/\lambda$, the MLE of mean wait is 2.5 hours. This illustrates invariance under a one-to-one parameter change.

- 10.3 Mean 4. Squared residual sum is $4 + 0 + 4 = 8$. MLE variance $8/3$; unbiased sample variance $8/(3 - 1) = 4$.
- 10.4 $\hat{\theta} = 7$. For $\theta \geq 7$, likelihood is θ^{-3} and decreases; for $\theta < 7$ it is zero because the observed value 7 lies outside the support.
- 10.5 $\hat{p} = n^{-1} \sum_i X_i$ has expectation p by linearity and variance $p(1 - p)/n$ by independence. Zero bias makes MSE equal variance.
- 10.6 For $0 \leq m \leq \theta$, $\mathbb{P}(M_n \leq m) = (m/\theta)^n$. Differentiating gives $f_M(m) = nm^{n-1}/\theta^n$. Integrating $mf_M(m)$ gives $\mathbb{E}M_n = n\theta/(n + 1)$. Therefore $(n + 1)M_n/n$ is unbiased.
- 10.7 The log-likelihood is a constant minus the positive coefficient $1/(2\sigma^2)$ times the residual sum of squares. Maximizing it is exactly minimizing that sum. This does not show squared error is appropriate for every noise model.
- 10.8 Right censoring contributes $\mathbb{P}_\lambda(X > 5) = e^{-5\lambda}$. A complete observation at 5 would contribute density $\lambda e^{-5\lambda}$. The extra factor λ changes the likelihood and incorrectly claims exact timing information that was never observed.

Bridge: the prerequisites for classes 22–23

This bridge is not a replacement for all of MIT classes 11–21. It supplies the concepts required by the selected confidence-interval classes: sampling distributions, estimation of variance, chi-square and t distributions, hypothesis testing, and the difference between confidence and posterior probability.

B1. Sampling design comes before formulas

An iid model is often an idealization of random sampling from a large stable population. Samples without replacement from a finite population are dependent; if the sampled fraction is appreciable, the standard error needs a finite-population correction. A convenience sample can systematically miss part of a population. Repeated requests from the same user may form clusters.

An interval quantifies the uncertainty represented by the model. It does not automatically include selection bias, sensor calibration errors, nonresponse, or changes in the process. Precise inference about the wrong target is still wrong.

B2. Why sample variance divides by $n - 1$

Define

$$S^2 = \frac{1}{n-1} \sum_i (X_i - \bar{X})^2, \quad n \geq 2.$$

The residuals sum to zero, so once $n - 1$ are known, the last is determined. That is the intuitive loss of one degree of freedom. Here is the actual unbiasedness proof, valid for iid finite-variance data, not just normal data:

$$\sum_i (X_i - \bar{X})^2 = \sum_i (X_i - \mu)^2 - n(\bar{X} - \mu)^2.$$

Taking expectations gives $n\sigma^2 - n(\sigma^2/n) = (n-1)\sigma^2$. Thus $\mathbb{E}[S^2] = \sigma^2$. Dividing instead by n gives expectation $(n-1)\sigma^2/n$. Although S^2 is unbiased for variance, $S = \sqrt{S^2}$ is generally not unbiased for SD.

B3. Chi-square: the distribution of squared normal noise

If $Z_1, \dots, Z_\nu$ are independent standard normals, define

$$V = \sum_{j=1}^{\nu} Z_j^2 \sim \chi_\nu^2.$$

It is nonnegative and right-skewed. Its mean is ν and variance 2ν . The mean follows from $\mathbb{E}Z^2 = 1$; the variance follows from $\mathbb{E}Z^4 = 3$, obtainable by integration by parts, so $\text{Var}(Z^2) = 3 - 1 = 2$.

For $v > 0$ its density is

$$f_V(v) = \frac{v^{\nu/2-1}e^{-v/2}}{2^{\nu/2}\Gamma(\nu/2)}, \quad \Gamma(a) = \int_0^\infty t^{a-1}e^{-t}dt.$$

To see the density's origin, integrate the rotationally symmetric joint normal density over spherical shells. A radius r has shell factor proportional to $r^{\nu-1}$. Set $v = r^2$, so $dr = dv/(2\sqrt{v})$; the resulting power is $v^{\nu/2-1}$. Normalizing by the gamma integral gives the displayed constant. This is a density derivation, not an assumption that chi-square itself is normal.

B4. Exact normal-sample geometry

For iid $X_i \sim \mathcal{N}(\mu, \sigma^2)$,

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1), \quad V = \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2,$$

and Z, V are independent. This is exact for normal data at any $n \geq 2$.

Proof through orthogonal coordinates. Standardize the vector of observations to $W_i = (X_i - \mu)/\sigma$. Its joint density is $(2\pi)^{-n/2}e^{-\|w\|^2/2}$. Rotate coordinates with an orthogonal matrix whose first row is $(1, \dots, 1)/\sqrt{n}$. Orthogonal transformations preserve length and have absolute determinant one, so the transformed density still factors as a product of independent standard normals. The first coordinate is Z ; the squared length of the other $n-1$ coordinates is

$$\sum_i W_i^2 - Z^2 = \frac{\sum_i (X_i - \bar{X})^2}{\sigma^2} = V.$$

It therefore has a chi-square law with $n-1$ degrees of freedom and is independent of the first coordinate. This proves the result and explains the degrees of freedom geometrically.

B5. Student t: a noisy denominator

Let $Z \sim \mathcal{N}(0, 1)$ and $V \sim \chi_\nu^2$ be independent. The ratio

$$T = \frac{Z}{\sqrt{V/\nu}}$$

defines the Student t distribution with ν degrees of freedom. For normal data the preceding result implies

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$

Replacing the true SD by an estimated SD adds randomness. Occasionally S is unusually small, making the ratio unusually large in magnitude. This gives t its heavier tails. With more degrees of freedom, V/ν concentrates near one, and t approaches standard normal.

For completeness, its density is

$$f_T(t) = \frac{\Gamma((\nu+1)/2)}{\sqrt{\nu\pi}\Gamma(\nu/2)} \left(1 + \frac{t^2}{\nu}\right)^{-(\nu+1)/2}.$$

Derive it by transforming (Z, V) to (T, V) using $z = t\sqrt{v/\nu}$. The Jacobian is $\sqrt{v/\nu}$. The integrand in v is proportional to $v^{(\nu+1)/2-1} \exp[-v(1+t^2/\nu)/2]$. Integrating with the gamma formula yields the displayed density. It is symmetric; its mean is zero for $\nu > 1$, and its variance is $\nu/(\nu-2)$ for $\nu > 2$. For smaller degrees these moments do not exist. Standardizing a sample statistic by S does not guarantee variance one.

B6. Hypothesis testing in the minimum necessary detail

A null hypothesis specifies a candidate model or parameter value, such as $H_0 : \mu = 100$. An alternative says what departures matter, such as $H_1 : \mu \neq 100$ or $H_1 : \mu > 100$. Choose a statistic and a rejection region whose probability under the null is at most a preselected significance level α .

A type I error rejects a true null. A type II error fails to reject a false null. Power is the probability of rejecting under a specified alternative. Failing to reject does not prove a null true; the data may simply be insufficiently informative.

For iid normal data with known σ , the two-sided z-test of $H_0 : \mu = \mu_0$ rejects when

$$\left| \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \right| > z_{1-\alpha/2}.$$

The p-value is the null-model probability of a test statistic at least as extreme as the observed one, according to the chosen test. Here it is $2[1 - \Phi(|z_{\text{obs}}|)]$. It is not $\mathbb{P}(H_0 \mid \text{data})$, nor the probability that the result was “due to chance.” For unknown σ and normal data, use the corresponding t statistic and t quantile.

B7. Bayesian intervals, only to prevent confusion

If we assign a prior density $\pi(\theta)$, Bayes gives

$$\pi(\theta \mid x) = \frac{L(\theta; x)\pi(\theta)}{\int L(u; x)\pi(u) du}.$$

A 95% credible interval contains 95% posterior probability under that prior and likelihood model. A frequentist confidence interval instead comes from a procedure with a stated repeated-sampling coverage. These are different guarantees, although some models and priors lead to numerically identical intervals.

For normal data with known variance and a flat prior on the mean, the posterior is normal with mean $\bar{x}$ and variance σ^2/n , so its central credible interval matches the usual z confidence interval. The flat prior on the whole real line is improper, but the resulting posterior is proper here. Equality of endpoints does not erase the distinct reasoning.

B8. Bridge checkpoint with answers

1. Why is $\bar{X} - \mu$ not a statistic if μ is unknown? Because its value cannot be computed from the observed sample and known constants alone.
2. With $n = 10$ normal observations, what are the degrees of freedom for the t statistic and standardized variance? Nine for both.
3. Does the exact t result require normal observations? Yes. For general iid finite-variance data, related large-sample approximations are available, but the finite-sample t identity is not automatic.
4. If $p = .03$, is there a 3% chance that the null is true? No. The p-value is a tail probability computed under the null model, not a posterior probability for the null.
5. Is an unbiased estimator guaranteed to equal the truth in a particular sample? No. Unbiasedness is an expectation over repeated samples.

Class 22: confidence intervals for normal data

22.1 Why this part exists

Two teams both report an average runtime of 100 ms. One measured four runs; the other measured ten thousand. Reporting only “100 ms” hides a huge difference in precision. A confidence interval combines an estimate, an uncertainty scale, and a coverage guarantee.

The central problem is not how to put brackets around a number. It is how to construct a rule that brackets the true parameter with a controlled frequency across repeated samples.

22.2 An interval is a statistic, not a fixed object before sampling

A confidence-interval procedure maps data to endpoints:

$$C(X) = [L(X), U(X)].$$

Both endpoints must be computable from data and known constants. If σ is unknown, $\bar{X} \pm 1.96\sigma/\sqrt{n}$ is not an implementable interval statistic. It contains an unknown input. Substituting S changes the statistic’s distribution and therefore changes its exact calibration.

For parameter θ and level $1 - \alpha$, a general coverage requirement is

$$\mathbb{P}_\theta\{\theta \in C(X)\} \geq 1 - \alpha \quad \text{for every allowed } \theta.$$

If equality holds for every θ , coverage is exactly $1 - \alpha$. The probability is over random data X , holding the unknown true parameter fixed. Coverage at only one convenient parameter value is not a uniform guarantee over the model.

A moving net and a fixed target

Think of the true parameter as a fixed target whose location you do not know. Each sample creates a new net, the interval. A 95% procedure catches the target on 95% of repeated throws under its assumptions. After one net has been thrown, that particular net either contains the target or does not. The 95% labels the procedure’s calibration, not a newly computed posterior probability of the target’s location.

Even among 100 independent repeats, the number of successful intervals need not be exactly 95. With exact independent 95% coverage it is binomial with mean 95 and variance $100(.95)(.05) = 4.75$.

22.3 Known variance: derive the z interval

Assume $X_i \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with σ known. Normality gives the exact pivot

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1).$$

A *pivot* may contain an unknown parameter, but its distribution does not depend on unknown parameters. This makes it useful for algebraically eliminating the unknown from the endpoints.

Let $c = z_{1-\alpha/2}$. Then

$$\mathbb{P}_\mu \left(-c \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq c \right) = 1 - \alpha.$$

Multiply by the positive standard error and rearrange:

$$-c \frac{\sigma}{\sqrt{n}} \leq \bar{X} - \mu \leq c \frac{\sigma}{\sqrt{n}} \iff \bar{X} - c \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + c \frac{\sigma}{\sqrt{n}}.$$

Thus the exact interval is

$$\boxed{\bar{x} \pm z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}}}.$$

This derivation is the proof of coverage. The familiar “estimate plus or minus margin” is the final form, not the starting axiom.

Worked example: calibrating an average measurement

Sixteen independent normal measurements have known population SD 8 units and observed mean 102. The SE is $8/4 = 2$. The 95% confidence interval for the population mean is $102 \pm 1.96(2) = [98.08, 105.92]$.

It is not an interval that contains 95% of individual measurements. Individual measurements fluctuate on scale 8, while the mean estimate fluctuates on scale 2. It also does not mean 95% of the observed data must lie in the interval.

The width tradeoff

Width is $2z_{1-\alpha/2}\sigma/\sqrt{n}$. Larger confidence means a larger critical value and a wider interval. More variation means a wider interval. More observations narrow it at the square-root rate. Halving the width requires four times the sample size, all else equal.

For a desired half-width $E > 0$, a known- σ design needs

$$n \geq \left(\frac{z_{1-\alpha/2}\sigma}{E} \right)^2,$$

rounded up to the next integer. A planning value for unknown σ is an assumption, not an observed certainty about future variability.

22.4 Unknown variance: derive the t interval

Under the same normal-data assumption, replace the known SD with sample SD. The bridge proved

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$

Choose $c = t_{1-\alpha/2, n-1}$ so $\mathbb{P}(-c \leq T \leq c) = 1 - \alpha$, then do the same algebra. The interval is

$$\boxed{\bar{x} \pm t_{1-\alpha/2, n-1} \frac{s}{\sqrt{n}}}.$$

For continuous nondegenerate normal sampling, $S > 0$ almost surely when $n \geq 2$, so multiplying by S preserves the inequalities. The larger t critical value accounts for uncertainty in estimating σ .

Worked example: same data summary, unknown SD

Let $n = 16$, $\bar{x} = 102$, and $s = 8$. Now the 95% t critical value is approximately 2.1314 with 15 degrees of freedom. The interval is

$$102 \pm 2.1314(8) \approx [97.737, 106.263].$$

The plug-in z interval would be narrower but would not have exact 95% finite-sample coverage. The difference matters most when n is small. There is no rule requiring you to stop using t once n exceeds 30; its quantiles naturally approach z quantiles.

22.5 Variance itself can be the target

In manufacturing or benchmarking, inconsistent performance may be the issue even if the mean is acceptable. For iid normal observations,

$$Q = \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2.$$

Let $a = \chi_{\alpha/2, n-1}^2$ and $b = \chi_{1-\alpha/2, n-1}^2$. Both are positive and $a < b$. Then

$$\mathbb{P}\left(a \leq \frac{(n-1)S^2}{\sigma^2} \leq b\right) = 1 - \alpha.$$

Inverting positive quantities reverses inequalities. Solving for the parameter yields

$$\left[\frac{(n-1)s^2}{\chi_{1-\alpha/2, n-1}^2}, \frac{(n-1)s^2}{\chi_{\alpha/2, n-1}^2} \right].$$

The larger chi-square quantile goes in the lower endpoint's denominator. The interval is generally asymmetric. Taking the square roots of both endpoints gives a confidence interval for σ , preserving coverage because square root is increasing.

Worked example: uncertainty about measurement variability

Ten independent normal measurements have sample variance $s^2 = 4$. With 9 degrees of freedom, the .025 and .975 chi-square quantiles are approximately 2.7004 and 19.0228. Thus

$$C_{\sigma^2} = \left[\frac{36}{19.0228}, \frac{36}{2.7004} \right] \approx [1.8925, 13.3314].$$

The SD interval is approximately $[1.3757, 3.6512]$. A sample SD of 2 is only a point estimate; ten observations leave substantial uncertainty about the population SD. Unlike a large-sample mean interval, this exact variance interval is particularly sensitive to the normality assumption.

If the population mean μ is genuinely known, $\sum_i (X_i - \mu)^2 / \sigma^2 \sim \chi_n^2$. The degrees of freedom become n because we did not estimate the center from the sample. Do not use this variant merely because a sample mean has been computed.

22.6 Connection with hypothesis tests

For a candidate null mean μ_0 , the two-sided z-test does not reject precisely when

$$\left| \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \right| \leq z_{1-\alpha/2}.$$

Rearranging shows this happens exactly when μ_0 belongs to the z confidence interval. The 95% interval is the collection of mean values not rejected by the corresponding 5% two-sided tests. The word “corresponding” matters: use matching models, sidedness, and calibrations.

In the known-SD example [98.08, 105.92], the null mean 100 is not rejected at 5%; the null mean 107 is rejected. Neither conclusion assigns posterior probabilities to these nulls. A wide interval can contain many incompatible practical possibilities simply because the data are imprecise.

22.7 One-sided bounds and prediction are different questions

If only an upper limit for the mean is needed, use

$$(-\infty, \bar{x} + z_{1-\alpha}\sigma/\sqrt{n}]$$

for known variance, or the corresponding t bound for unknown variance under normality. All of the allowed error α is placed in one tail, so 95% one-sided uses $z_{.95} = 1.6449$, not $z_{.975}$.

A prediction interval for an independent *future observation* needs to include both future individual noise and uncertainty in the estimated mean. For iid normal data with unknown variance,

$$\frac{X_{\text{new}} - \bar{X}}{S\sqrt{1 + 1/n}} \sim t_{n-1},$$

so the prediction interval is $\bar{x} \pm t_{1-\alpha/2, n-1}s\sqrt{1 + 1/n}$. To derive it, $X_{\text{new}} - \bar{X}$ is normal with variance $\sigma^2(1 + 1/n)$ and is independent of S ; then apply the t construction. This extension makes the distinction visible: a mean interval shrinks toward a point, but an individual-prediction interval retains the population’s inherent spread.

22.8 Practice: try before reading the solutions

- 22.1 With normal data, $n = 25$, $\bar{x} = 50$, and known $\sigma = 10$, find a 95% mean interval.
- 22.2 With normal data, $n = 10$, $\bar{x} = 12$, $s = 3$, use $t_{.975, 9} = 2.2622$ to find a 95% mean interval.
- 22.3 Known SD is 12. How many iid normal observations are needed for a 95% interval with half-width at most 2?
- 22.4 Explain whether $[\bar{X} - 2\sigma, \bar{X} + 2\sigma]$ is an interval statistic when σ is unknown.
- 22.5 For $n = 20$ normal observations with $s^2 = 9$, find a 95% variance interval using chi-square quantiles 8.9065 and 32.8523.
- 22.6 A 95% mean interval is [4.2, 5.8]. What do matching two-sided 5% tests say about null means 5 and 6? Does the interval prove the mean equals 5?
- 22.7 For $n = 100$, $\bar{x} = 20$, known $\sigma = 5$, find a 95% upper confidence bound for the mean.
- 22.8 Explain why $\bar{x} \pm 1.96s$ is generally neither the standard 95% mean interval nor the exact normal-data prediction interval.

22.9 Solutions and reasoning

- 22.1 SE is $10/5 = 2$. Interval $50 \pm 3.92 = [46.08, 53.92]$.
- 22.2 SE is $3/\sqrt{10} \approx .9487$; half-width about 2.1461; interval approximately [9.8539, 14.1461].

- 22.3 $n \geq (1.96 \cdot 12/2)^2 = 138.2976$, so 139 observations suffice.
- 22.4 No. The endpoints cannot be computed from observed data alone. If σ were known, it would be an interval statistic, although its coverage would depend on the sampling model and the chosen width.
- 22.5 Numerator $(n-1)s^2 = 171$. Lower endpoint $171/32.8523 \approx 5.2051$; upper $171/8.9065 \approx 19.1995$. The larger quantile belongs in the smaller endpoint.
- 22.6 Do not reject 5; reject 6. The interval does not select 5 as true. It expresses a set of parameter values compatible with the corresponding tests.
- 22.7 SE .5; upper endpoint $20 + 1.6449(.5) = 20.82245$. The confidence set is $(-\infty, 20.82245]$ unless the parameter space imposes a natural lower bound.
- 22.8 A mean interval needs an SE $s/\sqrt{n}$ and a t critical value for exact normal-data coverage. A prediction interval needs $s\sqrt{1 + 1/n}$ and a t critical value. The proposed expression omits the distinction between an estimated mean and an individual outcome.

Class 23: three views and non-normal data

23.1 Why this part exists

Normal data provide clean exact formulas. But polls are zeros and ones, runtimes can be skewed, and many models offer no convenient normal pivot. We need to understand the general construction of confidence intervals and identify when normal approximations remain justified.

This class connects the ideas: the pivot tells us how to build an interval algebraically; test inversion tells us how to build one from accept/reject rules; the formal definition tells us what property the result must actually satisfy.

23.2 View 1: start from a pivot and invert

Find $T(X, \theta)$ with a known distribution independent of unknown parameters. Choose a, b so that $\mathbb{P}_\theta(a \leq T(X, \theta) \leq b) = 1 - \alpha$. Solve those inequalities for θ . The resulting bounds must involve only data and known constants.

The three normal-data examples fit the same template:

Target	Pivot	Reference distribution
μ , known σ	$(\bar{X} - \mu)/(\sigma/\sqrt{n})$	Standard normal
μ , unknown σ	$(\bar{X} - \mu)/(S/\sqrt{n})$	t_{n-1}
σ^2 , unknown μ	$(n-1)S^2/\sigma^2$	χ_{n-1}^2

Choosing equal tails is conventional, not forced by the definition. A confidence procedure need not be symmetric, and shortest-length intervals may require different tail allocations.

23.3 View 2: invert an entire family of tests

For each candidate value θ_0 , suppose a level- α test has acceptance region $A(\theta_0)$ in data space. That means

$$\mathbb{P}_{\theta_0}(X \in A(\theta_0)) \geq 1 - \alpha.$$

After observing x , define

$$C(x) = \{\theta_0 : x \in A(\theta_0)\}.$$

This set collects all parameter values whose null hypotheses would not be rejected by the observed data.

Theorem 23.1 (Coverage from test inversion). *The set $C(X)$ has coverage at least $1 - \alpha$ for every allowed true parameter.*

Proof. For a fixed true value θ , the event $\theta \in C(X)$ is exactly the event $X \in A(\theta)$. Their probabilities are equal, and the latter is at least $1 - \alpha$ by the test's level guarantee. $\square$

The resulting object is formally a confidence *set*; in common one-dimensional problems it is an interval, but connectedness is not guaranteed for every model and every test family. Discrete tests often cannot attain exactly α without randomization, so coverage can exceed the nominal level.

Worked example: an exact non-normal interval from a maximum

For iid $X_i \sim \text{Unif}(0, \theta)$ let $M = \max_i X_i$. Its CDF is $\mathbb{P}_\theta(M \leq m) = (m/\theta)^n$ for $0 \leq m \leq \theta$. Hence

$$\mathbb{P}_\theta(M \geq \theta\alpha^{1/n}) = 1 - \alpha.$$

Since always $M \leq \theta$, inversion gives

$$C(X) = \left[M, \frac{M}{\alpha^{1/n}} \right].$$

It is an exact $(1 - \alpha)$ confidence interval without normal data, a CLT, or symmetry around an estimate. For $n = 5$, $M = 10$, and $\alpha = .05$, it is approximately $[10, 18.21]$. This is a strong reminder that “estimate ± 1.96 SE” is not the definition of confidence.

23.4 View 3: the formal coverage definition

All valid constructions must meet the coverage property. But coverage alone does not guarantee useful intervals. A data-independent randomized procedure could report the entire parameter space with probability .95 and the empty set with probability .05. It has 95% coverage yet offers little practical information.

Good procedures also aim for reasonable width, sensitivity to data, stability, and appropriate treatment of nuisance parameters. A nuisance parameter affects the sampling law but is not the main target, such as σ when estimating μ . Exact pivots are valuable because they remove that unknown from the reference distribution.

After observing an interval, a posterior probability that it contains θ requires a Bayesian model. Frequentist coverage by itself does not supply that posterior. Nor does 95% coverage mean every reported interval has the same conditional reliability after selecting only unusual data sets. Selection can change the operating properties.

23.5 Bernoulli proportions: the mean is a probability

Let $X_i \stackrel{\text{iid}}{\sim} \text{Bern}(p)$. The sample proportion is $\hat{p} = \bar{X} = k/n$. Its exact mean and variance are

$$\mathbb{E}[\hat{p}] = p, \quad \text{Var}(\hat{p}) = \frac{p(1-p)}{n}.$$

The CLT suggests

$$\frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \approx \mathcal{N}(0, 1),$$

for sufficiently informative samples with fixed $0 < p < 1$. The obstacle is that the denominator contains the unknown target p .

The conservative normal interval

Complete the square:

$$p(1-p) = \frac{1}{4} - \left(p - \frac{1}{2}\right)^2 \leq \frac{1}{4}.$$

Thus $\sqrt{p(1-p)/n} \leq 1/(2\sqrt{n})$. Using the largest possible SD produces

$$\hat{p} \pm \frac{z_{1-\alpha/2}}{2\sqrt{n}}.$$

Intersect with $[0, 1]$ because p cannot lie outside its parameter space. Intersecting with a known parameter space does not reduce coverage for admissible parameters.

Conservative does not remove approximation error

This interval is conservative relative to the CLT-based normal calculation because it overestimates the unknown SE. Its finite-sample binomial coverage is not automatically guaranteed to be at least the nominal level. A large-sample approximation is still being used. Exact binomial inversion or concentration bounds give genuine finite-sample guarantees under their assumptions.

At 95%, replace 1.96 by approximately 2 to get the memorable rule

$$\hat{p} \pm \frac{1}{\sqrt{n}}.$$

This is why random polls of about 1000 people often have margins near three percentage points under simple-sampling assumptions. Population size itself is not the main factor when the sampled fraction is small; sample size and sampling quality are.

Worked example: a product-feature poll

Among 400 independently sampled users from a target population, 232 prefer a new feature. Then $\hat{p} = .58$. The conservative normal 95% margin is $1.96/(2 \cdot 20) = .049$, giving $[\hat{p} - .049, \hat{p} + .049] = [.531, .629]$. The rule-of-thumb version is $[\hat{p} - .05, \hat{p} + .05] = [.53, .63]$.

That interval concerns the target population's preference proportion under the sampling model. If the survey only reached unusually active users, increasing n need not correct the selection bias. Also, a margin of .049 means 4.9 *percentage points*, not a 4.9% relative change from .58.

23.6 Planning the sample size

For a conservative normal proportion half-width E ,

$$n \geq \frac{z_{1-\alpha/2}^2}{4E^2}.$$

For nominal 95% and $E = .02$, this gives $n \geq 2401$ using 1.96. For $E = .03$, round 1067.11... upward to 1068. Halving a margin quadruples the sample size.

If a credible planning estimate p_0 is available, a less conservative formula uses $n \geq z_{1-\alpha/2}^2 p_0(1-p_0)/E^2$, but its adequacy depends on that planning assumption. The maximum-variance choice $p_0 = .5$ avoids underplanning due to an optimistic guessed proportion, though it does not fix CLT approximation or sampling bias.

23.7 Plug-in intervals and why boundaries are difficult

The familiar Wald interval substitutes the observed proportion in the SE:

$$\hat{p} \pm z_{1-\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

It can work for large samples with proportions away from zero and one, but coverage can be poor. If zero failures occur, it reports zero width around zero, incorrectly suggesting certainty. It can also extend outside $[0, 1]$. Simply clipping fixes impossible endpoints, not poor calibration.

Extension: Wilson interval via score-test inversion

Instead of estimating the denominator first, retain the candidate p inside the standard-error expression and solve

$$\frac{(\hat{p} - p)^2}{p(1-p)/n} \leq z^2.$$

Rearrangement gives the quadratic inequality

$$(n + z^2)p^2 - (2n\hat{p} + z^2)p + n\hat{p}^2 \leq 0.$$

Its roots yield the Wilson interval

$$\frac{\hat{p} + z^2/(2n) \pm z\sqrt{\hat{p}(1-\hat{p})/n + z^2/(4n^2)}}{1 + z^2/n}.$$

This interval is an extension beyond the selected core reading. It is still based on a normal-calibrated score test, not a universal exact guarantee, but it avoids the zero-width pathology at $\hat{p} = 0$ or 1 and often has better coverage than Wald.

Extension: exact binomial inversion

For observed $K = k$ from $\text{Bin}(n, p)$, one equal-tailed exact construction chooses a lower endpoint L solving $\mathbb{P}_L(K \geq k) = \alpha/2$ when $k > 0$, and an upper endpoint U solving $\mathbb{P}_U(K \leq k) = \alpha/2$ when $k < n$. Set $L = 0$ if $k = 0$ and $U = 1$ if $k = n$.

This is the Clopper-Pearson construction. Inverting binomial tails bounds the chance of excluding the true parameter on either side by $\alpha/2$; the union bound makes total noncoverage at most α . Discreteness usually causes overcoverage. “Exact” here means based on the exact sampling law with a guaranteed level, not guaranteed equality of coverage to the nominal number.

With zero failures, a one-sided exact upper 95% bound solves

$$(1-p)^n = .05 \implies p_U = 1 - .05^{1/n}.$$

For $n = 100$, $p_U \approx .02951$. Zero observed failures does not establish zero underlying risk. For large n , $p_U \approx -\log(.05)/n \approx 3/n$, the rule of three.

23.8 Non-normal sample means

Suppose observations are iid with finite mean μ and positive finite variance σ^2 , but not necessarily normal. The CLT gives

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1).$$

Also $S^2 \rightarrow \sigma^2$ in probability: write it using the empirical second moment and sample mean, then apply the LLN to X_i^2 and X_i . Since $\mathbb{E}X_i^2 < \infty$, those laws apply. Therefore $S/\sigma \rightarrow 1$ in probability.

Slutsky's theorem says multiplying a variable converging in distribution by one converging in probability to a constant multiplies the limiting distribution by that constant. It gives

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, 1).$$

Hence a large-sample approximate interval is

$$\boxed{\bar{x} \pm z_{1-\alpha/2} \frac{s}{\sqrt{n}}}.$$

Using a t critical value is also common and asymptotically equivalent, but it is not an exact t interval without the normal-data assumption. A large n rule must be judged against skewness, outliers, dependence, and the precision needed. Infinite variance defeats this reasoning.

Worked example: a skewed runtime distribution

Suppose 900 independent representative runtimes have observed mean 120 ms and sample SD 60 ms. If the finite-variance CLT approximation is adequate, SE is $60/30 = 2$ ms and the approximate 95% interval is $[116.08, 123.92]$ ms.

This does not claim runtimes are normal. It claims the estimated mean's sampling error is approximately normal. It also does not estimate a 95th-percentile runtime. A quantile requires a different statistic and uncertainty analysis.

23.9 A rigorous alternative: concentration for bounded data

This extension compares finite-sample guarantees with normal approximations. For iid Bernoulli data, Hoeffding's inequality states

$$\mathbb{P}(|\hat{p} - p| \geq \varepsilon) \leq 2e^{-2n\varepsilon^2}.$$

It yields the guaranteed interval

$$\left[\hat{p} - \sqrt{\frac{\log(2/\alpha)}{2n}}, \hat{p} + \sqrt{\frac{\log(2/\alpha)}{2n}} \right] \cap [0, 1].$$

It is usually wider than a nominal normal interval but does not require a large-sample normal approximation.

Proof for Bernoulli data. Let $g(t) = \log \mathbb{E}[e^{t(X-p)}]$. Then $g(0) = g'(0) = 0$. Differentiating shows $g''(t) = p_t(1 - p_t) \leq 1/4$, where $p_t = pe^t/(1 - p + pe^t)$. Taylor's theorem with an integral remainder yields $g(t) \leq t^2/8$ for all real t . For $t > 0$, Markov's inequality and independence give

$$\mathbb{P}\left(\sum_i (X_i - p) \geq n\varepsilon\right) \leq e^{-tn\varepsilon} \prod_i \mathbb{E}[e^{t(X_i - p)}] \leq \exp(-tn\varepsilon + nt^2/8).$$

Choosing $t = 4\varepsilon$ gives $e^{-2n\varepsilon^2}$. Apply the same argument to the lower tail and add the two bounds.

For error .02 and failure probability .05, Hoeffding requires $n \geq \log 40/[2(.02)^2]$, or 4612 observations after rounding. Compare 2401 from the conservative normal approximation and 12500 from Chebyshev. Each number carries a different mathematical guarantee.

23.10 Selection, repeated peeking, and multiple intervals

A fixed- n confidence guarantee does not automatically survive stopping whenever a preferred result appears. Likewise, constructing 20 separate 95% intervals does not make simultaneous 95% coverage for all 20 parameters. With independent exact procedures, simultaneous coverage would be $.95^{20} \approx .3585$.

One simple simultaneous guarantee uses Bonferroni: construct each of m intervals with noncoverage at most α/m . By the union bound, the chance that any misses is at most α . Independence is unnecessary for this bound. It may be conservative, but its logic is transparent.

23.11 Practice: try before reading the solutions

- 23.1 A random iid sample of 625 users contains 350 supporters. Find the conservative normal 95% interval and the rule-of-thumb version.
- 23.2 Plan a conservative normal 95% proportion interval of half-width .025. How many observations are needed?
- 23.3 An approximately CLT-valid sample has $n = 400$, $\bar{x} = 15$, $s = 8$. Find a large-sample 95% mean interval.
- 23.4 Explain why zero observed failures among ten trials does not justify the Wald interval $[0, 0]$.
- 23.5 For iid $\text{Unif}(0, \theta)$, $n = 4$ and observed maximum 8, find the exact 90% interval from the maximum construction.
- 23.6 Prove in one sentence why test inversion controls noncoverage by the test's type I error.
- 23.7 How many iid Bernoulli observations suffice, by Hoeffding, for error at most .05 with confidence at least .95?
- 23.8 Ten parameters need simultaneous coverage at least 95%. What per-interval confidence level suffices by Bonferroni?
- 23.9 A convenience survey has one million respondents. Its reported sampling margin is tiny. Does that establish negligible total estimation error? Explain.
- 23.10 Distinguish the statements: "95% of individual runtimes are below 200 ms," "a 95% CI for mean runtime ends at 200 ms," and "posterior probability that the mean is below 200 ms is .95."

23.12 Solutions and reasoning

- 23.1 $\hat{p} = 350/625 = .56$, $\sqrt{n} = 25$. Conservative normal margin $1.96/50 = .0392$, so $[.5208, .5992]$. Rule-of-thumb margin .04, so $[.52, .60]$.
- 23.2 $n \geq 1.96^2/[4(.025)^2] = 1536.64$, so 1537. Rounding down would miss the specified half-width target.
- 23.3 $SE\ 8/20 = .4$; half-width .784; interval $[14.216, 15.784]$. Its coverage is approximate and depends on sampling assumptions and CLT adequacy.
- 23.4 The empirical variance is zero but the true Bernoulli variance need not be. Many positive failure probabilities can produce ten successes. The one-sided exact 95% upper bound is $1 - .05^{1/10} \approx .2589$, showing how weak ten no-failure trials can be.

- 23.5 Here $\alpha = .1$, so the interval is $[8, 8/.1^{1/4}] \approx [8, 14.226]$. The parameter must at least equal the observed maximum, and the upper endpoint follows from its exact sampling law.
- 23.6 The true parameter is excluded from the confidence set exactly when the test rejects that same true parameter; that rejection is a type I error of probability at most α .
- 23.7 Require $2e^{-2n(.05)^2} \leq .05$, so $n \geq \log 40/.005 \approx 737.776$, hence 738.
- 23.8 Each noncoverage probability at most $.05/10 = .005$ suffices: use 99.5% marginal intervals. Summing the ten failure probabilities bounds total failure by .05.
- 23.9 No. A huge sample can estimate the wrong population's mean very precisely. Selection, nonresponse, and measurement biases are not represented by an iid sampling margin merely because n is large.
- 23.10 The first is a claim about the population CDF at 200. The second describes a repeated-sampling interval procedure for an unknown mean, and its sidedness must be specified. The third is a Bayesian posterior statement conditional on data, prior, and model. They target different quantities and probabilities.

Appendix A

Reference sheets and mathematical connections

A.1 Distribution selection sheet

Family	Mechanism / support	Mean	Variance
Bernoulli (p)	One yes/no result; $\{0, 1\}$	p	$p(1 - p)$
Binomial (n, p)	Fixed n iid yes/no trials; $0, \dots, n$	np	$np(1 - p)$
Geometric (p)	Trial index of first success; $1, 2, \dots$	$1/p$	$(1 - p)/p^2$
Uniform (a, b)	Constant density; $[a, b]$	$(a + b)/2$	$(b - a)^2/12$
Exponential (λ)	Constant hazard; $[0, \infty)$	$1/\lambda$	$1/\lambda^2$
Normal (μ, σ^2)	Additive noise model; $\mathbb{R}$	μ	σ^2
Pareto (m, α)	Power tail; $[m, \infty)$	$\alpha m/(\alpha - 1)$ if $\alpha > 1$	$\alpha m^2/[(\alpha - 1)^2(\alpha - 2)]$ if $\alpha > 2$
Chi-square (ν)	Sum of squared iid standard normals; $[0, \infty)$	ν	2ν
Student t (ν)	Normal divided by independent estimated scale; $\mathbb{R}$	0 if $\nu > 1$	$\nu/(\nu - 2)$ if $\nu > 2$

The table is a map, not permission to choose a distribution merely from a familiar name. Always connect the family to a mechanism or a checked statistical model. A normal approximation may be sensible for an average even when inappropriate for individual data.

A.2 Discrete uniform and sampling without replacement

For a uniform integer X on $\{1, \dots, m\}$, the PMF is $1/m$ at each point. The identities $\sum_{k=1}^m k = m(m + 1)/2$ and $\sum_{k=1}^m k^2 = m(m + 1)(2m + 1)/6$ give

$$\mathbb{E}X = \frac{m + 1}{2}, \quad \text{Var}(X) = \frac{m^2 - 1}{12}.$$

The first sum follows by pairing the first and last terms. The second follows by summing $(k + 1)^3 - k^3 = 3k^2 + 3k + 1$ and solving for the sum of squares. A fair six-sided die therefore has mean 3.5 and variance $35/12$.

For n draws without replacement from N items with M marked items, the number K of marked items is hypergeometric:

$$\mathbb{P}(K = k) = \frac{\binom{M}{k} \binom{N-M}{n-k}}{\binom{N}{n}}, \quad \max(0, n - N + M) \leq k \leq \min(n, M).$$

Count favorable unordered samples and divide by all samples. Let $p = M/N$. Then $\mathbb{E}K = np$ by indicators. For two different draw positions, the joint success probability is $M(M-1)/[N(N-1)]$, giving covariance $-p(1-p)/(N-1)$. Thus, for $N > 1$,

$$\text{Var}(K) = np(1-p)\frac{N-n}{N-1}.$$

The final factor is the finite-population correction. When $n = N$, every item is observed and the count has zero variance. This limiting case is a useful check on the formula.

A.3 Optional extension: Poisson counts and the exponential clock

Poisson is not a central required family in the selected early-class readings, but it helps connect count data with exponential waiting times. A Poisson variable has

$$\mathbb{P}(N = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, \dots, \quad \mathbb{E}N = \text{Var}(N) = \lambda.$$

Normalization is the exponential series. For its moments,

$$\mathbb{E}N = e^{-\lambda} \sum_{k \geq 1} \frac{k\lambda^k}{k!} = \lambda, \quad \mathbb{E}[N(N-1)] = \lambda^2,$$

giving variance $\lambda^2 + \lambda - \lambda^2 = \lambda$.

It arises as the rare-event limit of $\text{Bin}(n, \lambda/n)$: for fixed k ,

$$\binom{n}{k} (\lambda/n)^k (1 - \lambda/n)^{n-k} \rightarrow e^{-\lambda} \lambda^k / k!.$$

The falling-factorial term divided by n^k tends to one and the remaining exponential term tends to $e^{-\lambda}$.

In a homogeneous Poisson process of rate r , the number of arrivals by time t is $\text{Pois}(rt)$. The first arrival wait T exceeds t exactly when the count is zero, so $\mathbb{P}(T > t) = \mathbb{P}(N(t) = 0) = e^{-rt}$. Thus $T \sim \text{Exp}(r)$. Here r is a rate and rt is an expected count; distinguishing them prevents unit errors.

A.4 Optional extension: exact sum densities

For independent continuous X, Y , the density of $S = X + Y$ is the convolution

$$f_S(s) = \int_{-\infty}^{\infty} f_X(x) f_Y(s-x) dx.$$

Derivation: transform (X, Y) to (X, S) using $y = s - x$ and Jacobian one, then integrate out x . For two independent $\text{Unif}(0, 1)$ variables, the allowed x range has length s for $0 < s < 1$ and $2 - s$ for $1 < s < 2$, so the sum has a triangular density. Repeated convolution is one exact route behind the smoothing intuition of the CLT.

For independent normals, characteristic functions multiply:

$$e^{it\mu_X - \sigma_X^2 t^2/2} e^{it\mu_Y - \sigma_Y^2 t^2/2} = e^{it(\mu_X + \mu_Y) - (\sigma_X^2 + \sigma_Y^2)t^2/2}.$$

This is the characteristic function of a normal with summed means and variances. Thus sums of independent normals are exactly normal, not merely approximately so.

A.5 Empirical summaries versus population quantities

Given observed values $x_1, \dots, x_n$, the empirical CDF is

$$\hat{F}_n(t) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{\{x_i \leq t\}}.$$

It places mass $1/n$ on each observation, adding masses when values repeat. Its mean is $\bar{x}$. Its variance as a discrete distribution is $n^{-1} \sum_i (x_i - \bar{x})^2$, whereas the unbiased estimator of population variance is $s^2 = (n-1)^{-1} \sum_i (x_i - \bar{x})^2$. These are different mathematical objects, so different denominators are not a contradiction.

Sort the data as $x_{(1)} \leq \dots \leq x_{(n)}$. A sample median is the middle value for odd n ; averaging the middle two is a common convention for even n . Quantile interpolation conventions differ between software implementations. State the convention if exact sample quartiles matter. The population quantile definition through a CDF remains conceptually separate from an interpolation rule on a finite data set.

A.6 Confidence interval selection sheet

Situation	Interval / construction	Status and main assumptions
Mean; normal data; known SD	$\bar{x} \pm z_{1-\alpha/2} \sigma / \sqrt{n}$	Exact; iid normal; σ known.
Mean; normal data; unknown SD	$\bar{x} \pm t_{1-\alpha/2, n-1} s / \sqrt{n}$	Exact; iid normal.
Variance; normal data	Invert $(n-1)S^2/\sigma^2$ using chi-square quantiles	Exact; iid normal; strongly model-dependent.
Mean; non-normal data	$\bar{x} \pm z_{1-\alpha/2} s / \sqrt{n}$	Approximate; adequate iid finite-variance asymptotics.
Bernoulli proportion	$\hat{p} \pm z_{1-\alpha/2} / (2\sqrt{n})$	Conservative normal approximation, not universally exact.
Bernoulli; exact guarantee	Invert exact binomial tails	At least nominal coverage under iid Bernoulli sampling.
Bernoulli; simple rigorous bound	Hoeffding radius $\sqrt{\log(2/\alpha)/(2n)}$	At least nominal coverage; often wider.
Uniform upper endpoint	$[M, M/\alpha^{1/n}]$	Exact under iid $\text{Unif}(0, \theta)$.

A.7 A small normal table

z	0	.5	1	1.5	1.645	1.96	2
$\Phi(z)$	.5000	.6915	.8413	.9332	.9500	.9750	.9772
z	2.326	2.5	2.576	3			
$\Phi(z)$	.9900	.9938	.9950	.9987			

Use $\Phi(-z) = 1 - \Phi(z)$ for negative values. The number .975 for $z = 1.96$ is a *left-tail* probability; subtracting the .025 left tail gives .95 in the central region.

A.8 Proof and assumption audit

- Counting identities: finite combinatorial proofs in class 1.
- Probability rules, conditional multiplication, total probability, Bayes: derived from axioms and definitions in classes 2–3.
- Linearity and indicator expectations: proved without independence in class 4.
- Variance identities and family moments: proved in classes 5–6.
- Transformation formula, normal normalization, exponential memorylessness: derived in class 5.
- Markov, Chebyshev, finite-variance weak LLN, histogram convergence: proved in class 6.
- CLT: characteristic-function proof in class 6, importing the explicitly stated continuity theorem; general strong LLN stated but not proved.
- Covariance identities, correlation bounds, total expectation and variance: proved in class 7.
- MLE formulas and bias/MSE facts: derived in class 10 and its exercises.
- Normal sample mean/variance independence and chi-square law: orthogonal-coordinate proof in the bridge; t density obtained by transformation.
- Confidence intervals: coverage derived by pivots and test inversion in classes 22–23.
- Approximate non-normal intervals: CLT, consistency of S , and the explicitly stated Slutsky theorem.
- Hoeffding bound for Bernoulli data: proved in class 23 as an optional finite-sample extension.

This audit identifies the mathematical dependencies. A theorem's assumptions are part of the theorem, not a footnote to be dropped when applying it.

Appendix B

Four computational studios

MIT 18.05 uses R for computation and simulation. These studios give small base-R examples and precise questions. They require no downloaded data or specialist package. Simulation illustrates theory; it does not replace proof. Random results vary unless the seed and software behavior are held fixed.

B.1 R notation in one page

Assignment uses `<-`. `c(1,2,3)` constructs a vector. `mean(x)`, `sd(x)`, and `var(x)` compute summaries; `var` uses the sample-variance denominator $n-1$. `replicate(B, expression)` repeats an expression B times. Logical values behave as zeros and ones when averaged, so `mean(x > 3)` is a fraction.

Distribution functions use four prefixes: **d** for PMF or density, **p** for CDF, **q** for quantile, and **r** for random sampling. For example, `dnorm`, `pnorm`, `qnorm`, and `rnorm`. Normal functions use `sd`, not variance; exponential functions use `rate`, not mean. R's `rgeom` counts failures before success, so add one for this book's trial-count convention.

```
dbinom(3, size=10, prob=0.2) # P(K = 3)
pbinom(3, size=10, prob=0.2) # P(K <= 3)
pnorm(130, mean=100, sd=15, lower.tail=FALSE)
qnorm(0.975) # 1.959964...
qt(0.975, df=15) # t critical value
qchisq(c(0.025,0.975), df=9)
rexp(100, rate=2) # mean 1/2, not 2
1 + rgeom(100, prob=0.2) # trials until success
```

B.2 Studio 1: one path of averages versus many averages

Task. Simulate 5000 fair bits and plot the running mean. Then independently repeat an experiment of 100 fair bits 10000 times and inspect the distribution of those 100-bit averages. Explain why the two plots answer different questions.

```
set.seed(1805)
x <- rbinom(5000, size=1, prob=0.5)
running <- cumsum(x) / seq_along(x)
plot(running, type="l", xlab="Number of trials",
     ylab="Running proportion")
abline(h=0.5, col="red")

batch_means <- replicate(10000, {
  mean(rbinom(100, size=1, prob=0.5))
})
hist(batch_means, probability=TRUE, breaks=30)
```

```
mean(batch_means)
sd(batch_means)
```

Solution and interpretation. The first graph is one evolving sample path and illustrates LLN stabilization with temporary reversals. The second approximates a sampling distribution at fixed $n = 100$. Its mean should be near .5 and SD near $\sqrt{.25/100} = .05$. The CLT predicts an approximately bell-shaped, still discrete distribution of batch averages. There is no theorem that the first running path must move monotonically closer to .5.

B.3 Studio 2: a skewed population can have a near-normal average

Task. For $n = 1, 5, 30, 100$, generate many means of iid exponential variables of rate one. Plot both raw means and standardized means. Check the predicted SDs.

```
set.seed(1806)
for (n in c(1,5,30,100)) {
  means <- replicate(10000, mean(rexp(n, rate=1)))
  z <- sqrt(n) * (means - 1)
  print(c(n=n, mean=mean(means), sd=sd(means)))
  hist(z, probability=TRUE, breaks=50,
       main=paste("Standardized means, n =", n))
  curve(dnorm(x), add=TRUE, col="red", lwd=2)
}
```

Solution and interpretation. Population mean and SD are both one. The raw mean should have mean near one and SD near $1/\sqrt{n}$. The standardized mean should have mean near zero and SD near one for each n , while its shape becomes more normal as n grows. Mean zero and variance one alone are not enough to make a distribution normal; the changing shape is the additional CLT statement.

B.4 Studio 3: confidence is a property of repetitions

Task. Repeatedly sample $n = 10$ normal observations with true mean 5 and SD 2. Compare exact t intervals with intervals that incorrectly use 1.96 and estimated SD.

```
set.seed(1822)
B <- 20000
n <- 10
covered <- replicate(B, {
  x <- rnorm(n, mean=5, sd=2)
  center <- mean(x)
  se <- sd(x) / sqrt(n)
  c(t_interval=abs(center-5) <= qt(0.975,n-1)*se,
    plug_in_z=abs(center-5) <= qnorm(0.975)*se)
})
rowMeans(covered)
```

Solution and interpretation. The t procedure has exact 95% model coverage. Simulation should be close, not identical, to .95. With $B = 20000$, a simulated coverage fraction near .95 has Monte Carlo SE about $\sqrt{.95(.05)/20000} = .00154$. The plug-in z method undercovers because its

critical value is too small for the t-distributed statistic. Its exact coverage here is $\mathbb{P}(|t_9| \leq 1.96)$, about .9184.

B.5 Studio 4: a dependence trap

Task. Generate $X_i = U + \varepsilon_i$ with $U \sim \mathcal{N}(0, 4)$ and independent $\varepsilon_i \sim \mathcal{N}(0, 9)$. Each batch gets one fresh shared U . Compare empirical variance of batch means with the naive iid calculation.

```
set.seed(1807)
n <- 100
means <- replicate(20000, {
  shared <- rnorm(1, mean=0, sd=2)
  errors <- rnorm(n, mean=0, sd=3)
  mean(shared + errors)
})
var(means)
4 + 9/n      # correct variance of average
(4 + 9)/n    # wrong if independence is assumed
```

Solution and interpretation. The variance of one observation is 13, but the observations within a batch are not independent. Their covariance is 4. The correct average variance is 4.09, not .13. The experiment isolates a practical failure mode: pretending that repeated measurements under the same shared condition are independent can grossly overstate precision.

Appendix C

Six synthesis problems with solutions

These additional problems require you to decide what tool to use. They are intentionally not organized by formula. Together with the four studios and the 90 numbered class/exam questions, they make 100 practice tasks, in addition to the bridge checkpoints.

C.1 Problems

- C1 **No observed failures.** A new service completes 200 independent representative requests without failure. (a) Find the MLE of failure probability. (b) Find an exact one-sided 95% upper bound. (c) Explain why these answers are compatible rather than contradictory.
- C2 **Randomized search.** Each attempt independently finds an acceptable result with probability .05. Each attempt costs 4 ms. (a) Find expected time and its SD. (b) Find the smallest number of attempts giving at least 95% success probability. (c) Explain why an expected-time budget is not a 95% budget.
- C3 **Two benchmark regimes.** A job runs on a quiet machine with probability .8 and a busy one with probability .2. Quiet runtimes have mean 10 and variance 4; busy runtimes have mean 30 and variance 9. Find the overall mean and variance. State one reason a single normal model may be misleading.
- C4 **Learning an exponential clock.** Ten complete iid waits sum to 25 seconds. (a) Estimate the rate by MLE. (b) Give the estimated probability of waiting longer than 5 seconds for a new event. (c) Explain which uncertainty this probability ignores. (d) If instead the last wait was censored after 5 seconds, with the first nine complete waits summing to 20, derive the new rate MLE.
- C5 **How many simulation trials?** A Monte Carlo experiment estimates a probability using independent indicators. (a) Use the conservative normal approximation for error .01 at nominal 95%. (b) Use Hoeffding for a finite-sample guarantee at the same levels. (c) Explain why neither result detects a bug in the simulated model.
- C6 **Compare two independent means, extension.** Two independent normal samples have known population SDs 10 and 12, sizes 100 and 144, and observed means 50 and 47. Derive an exact 95% interval for the difference of population means. Would a matched-pairs design with positive covariance use the same standard error?

C.2 Solutions

- C1 The Bernoulli MLE is zero. The exact one-sided upper bound solves $(1 - p)^{200} = .05$, giving $1 - .05^{1/200} \approx .01487$. The likelihood peaks at zero, but many small positive probabilities remain compatible with the observations. A point estimate need not convey the uncertainty in that estimate. Independence and representativeness of requests are essential assumptions.

- C2 Trial count T has mean 20 and variance $.95/ (.05)^2 = 380$, so time mean is 80 ms and time SD is $4\sqrt{380} \approx 77.974$ ms. Require $1 - .95^k \geq .95$, equivalently $k \geq \log(.05)/\log(.95) \approx 58.404$. Thus 59 attempts, or 236 ms, achieves at least 95% success under the model. A large spread explains the difference from the 80 ms mean.
- C3 Total expectation gives $.8(10) + .2(30) = 14$. Average conditional variance is $.8(4) + .2(9) = 5$. Variance of conditional means is $.8(10 - 14)^2 + .2(30 - 14)^2 = 64$. Total variance is 69. A mixture of two regimes can have structure, skewness, or two peaks that a normal model centered at 14 conceals.
- C4 (a) $\hat{\lambda} = 10/25 = .4$ per second. (b) Plug-in survival probability $e^{-.4(5)} = e^{-2} \approx .1353$. (c) This treats the estimated rate as though known and ignores parameter-estimation uncertainty and possible model misspecification. (d) Nine exact waits contribute $\lambda^9 e^{-20\lambda}$ and one censored wait contributes $e^{-5\lambda}$. Log-likelihood is $9 \log \lambda - 25\lambda$, so $\hat{\lambda} = 9/25 = .36$, not .4.
- C5 (a) $n \geq 1.96^2/[4(.01)^2] = 9604$. (b) $n \geq \log 40/[2(.01)^2] \approx 18444.397$, so 18445. The second is a conservative non-asymptotic guarantee. (c) Both concern sampling error *inside* the implemented model. A biased random-number procedure or a wrong event definition can produce a precise answer to the wrong problem.
- C6 The difference of sample means is normal with mean $\mu_1 - \mu_2$ and variance $10^2/100 + 12^2/144 = 2$. The observed difference is 3, giving $3 \pm 1.96\sqrt{2} \approx [.228, 5.772]$. For paired observations, the variance of differences includes a negative $2 \text{Cov}(X, Y)$ term. Positive pairing covariance can reduce the SE; the independent-samples formula is then not the correct design-specific expression.

Appendix D

Sources, scope, and your next study pass

D.1 Official MIT alignment

The class numbering and topic selection in this companion follow the MIT OpenCourseWare offering *18.05 Introduction to Probability and Statistics, Spring 2022*. The OCW course lists Jeremy Orloff and Jennifer French Kamrin as instructors; the reading PDFs identify Jeremy Orloff and Jonathan Bloom as authors. The companion’s explanatory prose, examples, exercises, and organization within chapters are original educational material, not an official course transcript or an endorsement by MIT.

- [MIT class-by-class readings and in-class materials](#): authoritative scope map, including the split class 5 and class 6 readings.
- [MIT course calendar](#): confirms class 8 as review and class 9 as Exam 1.
- [MIT combined probability readings](#): selected early-class source reference, including the histogram and Chebyshev appendix.
- [MIT class 10b: maximum likelihood](#): reference for the class’s estimation scope.
- [MIT class 22: confidence intervals based on normal data](#): reference for z , t , chi-square, and test connections.
- [MIT class 23a: three views of confidence intervals](#): reference for pivotal, test-inversion, and formal coverage viewpoints.
- [MIT class 23b: means of non-normal data](#): reference for conservative normal proportion intervals and large-sample means.

D.2 What was deliberately added

The original practice exam, software-system analogies, inverse-CDF sampling discussion, explicit CLT proof route, normal-sample geometry proof, Wilson and exact-binomial comparisons, Hoeffding proof, censoring example, prediction interval, and synthesis problems provide enrichment. They are included to make the selected classes more understandable and self-contained, not because every one is an official lecture requirement.

This book does not claim to cover the unrequested semester topics in full. In particular, the Bayesian updating sequence, the entire hypothesis-testing sequence, bootstrap confidence intervals, and regression need their own study. The bridge is only what is necessary to understand the requested confidence-interval material.

D.3 A final understanding test

Before calling a topic mastered, answer these questions in ordinary language:

1. What is random in this problem, and what is fixed but unknown?
2. What event, parameter, or statistic am I actually trying to compute?
3. Why is the selected model appropriate for the data-generating mechanism?
4. Which assumptions justify independence, a density, a normal approximation, or a particular reference distribution?
5. Is my answer exact, approximate, or a bound?
6. What would make it misleading in real life even if the algebra is correct?

If you can answer those six questions and solve the exercises without guessing a formula, you have learned more than a list of distributions. You have learned how probability models uncertainty and how statistics turns observations into carefully qualified conclusions.